

# Sharp Target-Domain Certificates for Quantum-Kernel Advantage under Distribution Shift

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## Abstract

Quantum predictive advantage under shift usually requires known target labels. We derive the assumption-free sharp identified interval for the finite-batch advantage of a fixed candidate over the best member of a prespecified fixed classical-kernel family under any bounded loss and unrestricted completions of the unaudited labels; zero-one accuracy has exact closed-form partial-label updates. Across eight security shifts, zero-label upper endpoints against 115 classical kernels span 0.002–0.088. Retrospective auditing reduces every endpoint to 0.010 with 0–33 of 500 labels, including entangling-ZZ models. Prospective corroboration on two eligible tasks met predefined criteria: four task-classifier medians require 0–3 labels (overall median 0; maximum 76), and all 20 realized effects are negative; a third task fails its feature gate. Protocol controls reverse an apparent advantage, while finite-shot noise can increase predictive distinctness without useful advantage. The framework separates protocol validity, target predictive indispensability, and quantum relevance.

**Keywords:** quantum machine learning; quantum kernels; quantum advantage; partial identification; active testing; distribution shift; classical surrogates; kernel methods

# 047 1 Introduction

048  
049 Quantum kernel methods embed classical data into quantum states and use state  
050 overlaps as similarities for otherwise classical learners [1–3]. This separation makes  
051 them a clean setting for studying quantum inductive bias, but a predictive-advantage  
052 claim must pass three distinct tests. The comparison protocol must be valid; the target  
053 predictions must resist reproduction by an appropriate classical reference family; and  
054 the surviving behaviour must depend on a non-trivial quantum construction whose  
055 resources are counted. A model can pass one test and fail another.

061 This article therefore studies a deliberately narrow object: for a fixed quantum  
062 candidate and a prespecified classical-kernel reference family, the set of predictive  
063 advantages on the actual deployment batch that remain compatible with the tar-  
064 get labels not yet inspected. We make four contributions. First, for any additive  
065 bounded loss on a finite label space, we derive the interval hull of the partial-label  
066 identified set for advantage over the best member of a prespecified classical-kernel  
067 reference family and show that no assumption-free rule using the same information  
068 can return a uniformly smaller valid interval. Zero-one accuracy has an unusually  
069 simple closed form and is pathwise exact after any realized audit subset, including  
070 subsets selected adaptively. Second, nested classical reference families and target-  
071 label audits define a reference-breadth-target-supervision evidence frontier. Third, we  
072 give an exact prevalence-conditional balanced-accuracy formulation and an explicit  
073 PSD construction showing why train-only geometry cannot determine target predic-  
074 tions. Fourth, we embed these objects in a controlled quantum-kernel benchmark  
075 and prospectively specify a Gate 2 replication before opening target labels; entan-  
076 glement strata and repeated finite-shot perturbations separately test when predictive  
077 distinctness is quantum-relevant.

089 The benchmark uses four security scenarios from EMBER, UNSW-NB15, and ToN-  
090 IoT, producing eight fixed constructed or held-out-campaign shifts. Support-vector  
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and Laplace Gaussian-process classifiers consume identical precomputed Gram matrices. The quantum family contains entangling-ZZ and separable-product fidelity maps at 4–12 qubits; the classical family contains 115 linear, RBF, polynomial, Laplacian, and Matérn candidates. Models are tuned on training data and selected on a disjoint ID-validation split (Figure 1). Equal 60-candidate pools support fair performance comparison, while the complete prespecified classical-kernel reference family supports the separate Gate 2 indispensability audit. One prospectively specified three-task TableShift study corroborates protocol sensitivity; a second tests Gate 2 prospectively on previously unused task identifiers.

Against the full prespecified classical-kernel reference family, the zero-label upper endpoint of accuracy advantage ranges from 0.002 to 0.050 for SVC and 0.006 to 0.088 for GPC. In a retrospective fixed-batch analysis, an adaptive bottleneck-coverage audit reduces it to at most 0.010 using 0–33 of 500 labels (median 13 over 16 fixed case-classifier cells). Dynamic random-disagreement and non-adaptive initial-coverage controls both have median 12.5, so the large gain over prediction-independent ordering is attributable to excluding non-actionable queries rather than to the proposed coverage heuristic. The result persists for selected entangling-ZZ models. Prospective corroboration on two technically eligible tasks met the predefined criteria: four task-classifier medians were 0–3 labels, the overall seed-level median was 0, and the maximum was 76. All 20 prospective realized accuracy effects were negative. The third task failed its prespecified minimum-feature gate before model execution. The certificate remains conditional on the prespecified classical-kernel reference family and target batch, not evidence of classical simulability or complexity-theoretic separation. It converts the earlier benchmark conclusion—that controls remove an apparent gain—into a quantitative question: how much advantage remains possible, and how much classical search and target supervision suffice to falsify a material claim?

## 139 1.1 Relation to prior work

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141 Recent benchmarks emphasize the first test. Broad QML comparisons find competi-  
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143 tive classical baselines once pipelines are aligned [4]; Slattery et al. show numerically  
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145 that general-purpose hyperparameter tuning can make fidelity kernels well approxi-  
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147 mated by classical kernels on classical data [5]; and quantum-kernel results depend  
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149 strongly on feature maps and hyperparameters [6]; and a recent intrusion-detection  
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151 audit with tuned classical baselines, calibration-aware metrics, and a simulated noise  
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153 sweep finds that tuned classical models match or exceed hybrid and quantum-kernel  
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155 models on aggregate detection, with only two narrow exceptions surviving multiplicity  
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157 correction [7]. Those studies concern benchmark performance or global kernel approx-  
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159 imation. Our contribution instead gives an exact finite-target interval conditional on  
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161 fixed predictions and a prespecified classical-kernel reference family, including under  
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163 partial target-label auditing. A concurrent finance preprint likewise shows an appar-  
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165 ent quantum-kernel advantage vanishing under point-in-time, kernel-swap, and budget  
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167 controls [8]. Under distribution shift the selection rule becomes part of the estimand:  
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169 same-test selection is an oracle, train-only regularization can change rankings, and  
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171 a larger candidate family has more opportunities to win [9–12]. Fair equal-budget  
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173 benchmarking answers who wins under symmetric search, but not whether any clas-  
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175 sical witness in a declared family can reproduce the selected quantum model on the  
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177 deployment inputs.

178 Geometric and surrogate analyses approach that second question from complemen-  
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180 tary directions. Huang et al. use a label-independent geometric difference to screen  
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182 whether a quantum kernel can have a prediction advantage [13]; classical-surrogate  
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184 constructions reproduce input–output relations of trained quantum models [14]; a  
closed-form classical surrogate has been derived for the ZZ feature map in the small-  
bandwidth regime [15]; and recent theory studies when quantum learners depart from  
minimum-norm classical solutions [16]. Those objects characterize geometric potential,

global approximation, or learning-rule separation. Our target-batch question is different. A train-only Gram matrix does not determine how a kernel transports predictions to a shifted target because deployment also depends on the target–train block, while different geometries can yield identical decisions on the observed target inputs. We therefore audit predictive indispensability directly on target covariates while keeping their labels protected.

That protection creates an identification problem in the tradition of partial identification: observed information and maintained restrictions can determine a set of compatible values without point-identifying the target [17, 18]. Here the observed object is unusually concrete—one finite batch and fixed prediction matrices—and the maintained label restriction is deliberately empty. The estimand is the relative functional “fixed candidate versus best member of a prespecified fixed family.” We derive its sharp interval over unrestricted completions of the unaudited labels, rather than estimate a population parameter or add assumptions until point identification is recovered. Weak-supervision work also uses partial identification, but combines fitted label-model marginals with Fréchet restrictions for individual metrics [19]; those marginals are neither available nor required here.

Label-efficient evaluation pursues a different objective. Active evaluation has used adaptive stratified sampling [20], importance-weighted Active Testing [21], learned loss surrogates [22], and prediction-powered control variates [23] to estimate risk accurately with fewer labels. Limited-label model selection instead chooses which pretrained classifier to deploy [24], while semi-supervised multi-model evaluation borrows information across classifier scores and unlabeled data under a mixture model [25]. Under distribution shift, ATC predicts target accuracy from confidence [26]; disagreement-discrepancy methods provide assumption-dependent error bounds [27, 28]; and label-complexity results show that, under joint covariate–label shift, no label-free method is simultaneously per-class valid and efficient uniformly over the

target laws consistent with the source data and target covariates [29]. Our certificate estimates neither labels nor risk: it retains every completion and reports the exact range of a fixed relative functional. Co-validation and discordant-pair evaluation remain complementary uses of disagreement [30, 31].

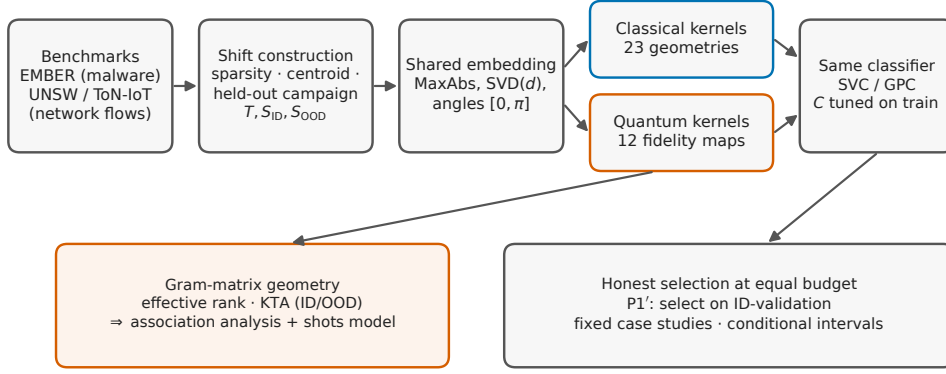
The distinction from minimax transductive learning is equally important. Balsubramani and Freund formulate ensemble prediction on an unlabeled test set as a game, using prior classifier-label correlation constraints to choose a new predictor that minimizes worst-case error [32]. We do not optimize or combine predictions and impose no such label constraints. The candidate and reference predictions remain fixed; the adversarial completion is used only to identify the sharp range of their realized finite-batch advantage. Thus transductive minimax aggregation is a decision rule under supplied performance information, whereas our object is an assumption-free identification certificate conditional on a declared prediction matrix. The distinction concerns the target, admissible information, and action, not only the computational form.

A companion study by the present authors addresses a different question in a different domain: the conditional validity of quantum event classifiers under collider systematics and finite-shot estimation uncertainty, using an information-conditional, fail-closed auditing framework with anytime-valid per-claim error control [33]. It shares neither datasets, estimands, nor guarantees with the finite-batch certificate developed here.

## 2 Results

### 2.1 Overview

The results are organized as three gates. Gate 1 asks whether the comparison protocol itself is valid: regularization is fitted on training data, deployment candidates are selected without OOD labels, and family search budgets are distinguished from unrestricted adversarial reference breadth. Gate 2 asks whether the selected quantum



**Fig. 1** The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by target-label-free, equal-budget selection on an in-distribution validation split (P1'). Each scenario-group interval is a 95% conditional cluster- $t$  interval over five q-split seed clusters ( $n = 5$ ), conditional on the fixed benchmark pools and design. Gram matrices feed the associational geometry audit and repeated finite-shot sensitivity.

model is predictively indispensable relative to the prespecified classical-kernel reference family on the observed target inputs. This is the new target-batch certificate and evidence frontier. Gate 3 asks whether any surviving behaviour is quantum-relevant, separating entangling-ZZ maps from product maps and accounting for finite-shot estimation. The eight security groups are fixed cases; conditional intervals over q-split clusters describe pipeline variation given the benchmark pools and design, not uncertainty over a population of datasets. Extended results, complete sensitivity tables, and diagnostics are provided as Supplementary Information (Additional file 1).

## 2.2 Sharp target-batch identification of predictive advantage

Consider a finite target batch, a fixed quantum prediction, and  $M$  fixed classical predictions. For a finite label space  $\mathcal{Y}$  and an additive bounded loss, let  $\ell_i^Q(y)$  and  $\ell_{ij}^C(y)$  be their losses on item  $i$  if its label were  $y$ , and define the classical-minus-quantum

323 contrast  $a_{ij}(y) = \ell_{ij}^C(y) - \ell_i^Q(y)$ . Positive advantage favours the quantum model:  
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$$\Delta_\ell(y) = \min_{j \leq M} \frac{1}{n} \sum_{i=1}^n a_{ij}(y_i).$$

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 330 If labels have been audited on  $L$  and  $S_j(L) = \sum_{i \in L} a_{ij}(y_i)$ , every assignment on  $L^c$   
 331 remains admissible.  
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333  
 334 **Theorem 1** (Sharp bounded-loss interval hull: sharpness and minimality) *The smallest*  
 335 *interval containing  $\Delta_\ell(y)$  for every completion of the unaudited labels is  $[L_\ell(L), U_\ell(L)]$ , where*  
 336  
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$$L_\ell(L) = \frac{1}{n} \min_j \left\{ S_j(L) + \sum_{i \notin L} \min_{y \in \mathcal{Y}} a_{ij}(y) \right\},$$

$$U_\ell(L) = \frac{1}{n} \max_{y_{L^c} \in \mathcal{Y}^{|L^c|}} \min_j \left\{ S_j(L) + \sum_{i \notin L} a_{ij}(y_i) \right\}.$$

341  
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 344 *Both endpoints are attained. Consequently, any assumption-free interval computed from the*  
 345 *same fixed losses and audited labels and valid for every admissible completion must contain*  
 346 *both endpoints.*  
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 350 The lower endpoint separates over labels after choosing a witness. The upper  
 351 endpoint is an exact finite max-min problem, expressible as a mixed-integer linear  
 352 program with one-hot label variables. Auditing any additional label removes comple-  
 353 tions, so the lower endpoint cannot decrease and the upper endpoint cannot increase.  
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 355 This general statement is not a claim that loss-based partial identification or active  
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 357 evaluation is new; the contribution is the exact relative-family estimand and its sharp,  
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 359 minimal finite-batch certificate. Zero-one classification has further structure: assign-  
 360 ing an unaudited label to the quantum hard prediction simultaneously maximizes its  
 361 contrast with every disagreeing classical witness. This yields the following closed forms  
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 363 used in all experiments.  
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Fix the hard target predictions of a selected quantum classifier  $h_Q$  and a pre-specified classical family  $\mathcal{C}_B = \{h_{C_1}, \dots, h_{C_M}\}$  on  $n$  target inputs. For an otherwise unrestricted target labelling  $y$ , define

$$\Delta_B(y) = \text{Acc}(h_Q; y) - \max_{j \leq M} \text{Acc}(h_{C_j}; y), \quad d_j = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{h_Q(x_i) \neq h_{C_j}(x_i)\}.$$

**Theorem 2** (Sharp zero-label region) *The smallest interval containing  $\Delta_B(y)$  for every possible target labelling is*

$$\mathcal{I}_B^{(0)} = \left[ -\max_{j \leq M} d_j, \min_{j \leq M} d_j \right].$$

*Both endpoints are attainable, including for multiclass hard predictions.*

The upper endpoint is not a probabilistic bound or an estimate: it is the exact largest advantage compatible with the observed prediction matrix when every target label remains unknown. It is attained by setting  $y = h_Q$ . The lower endpoint is attained by assigning the predictions of the classical model farthest from  $h_Q$ . Hence no smaller label-free interval is valid without extra assumptions. In particular, if  $\min_j d_j < \tau$ , any accuracy advantage of at least  $\tau$  over this classical family is falsified on the batch, even before a target label is revealed. Whenever  $\min_j d_j > 0$ , the same target covariates and prediction matrix remain compatible with both signs of advantage; disagreement alone cannot identify which family wins.

The zero-label certificate is already restrictive in several cases (Figure 2a). Against the complete 115-candidate classical family, the upper endpoint ranges from 0.002 to 0.050 for SVC and from 0.006 to 0.088 for GPC (Table 1). A small endpoint says that at least one classical witness nearly reproduces the selected quantum decisions; it does not say that the quantum kernel is efficiently simulable or globally reproducible.

**Theorem 3** (Sharp arbitrary-partial-label region) *Let  $L$  be any audited subset. For classical witness  $j$ , let*

$$S_j(L) = \sum_{i \in L} [\mathbf{1}\{h_Q(x_i) = y_i\} - \mathbf{1}\{h_{C_j}(x_i) = y_i\}]$$

*and let  $R_j(L)$  count unaudited points on which  $h_Q$  and  $h_{C_j}$  disagree. Then the sharp identified interval over all completions of the unaudited labels is*

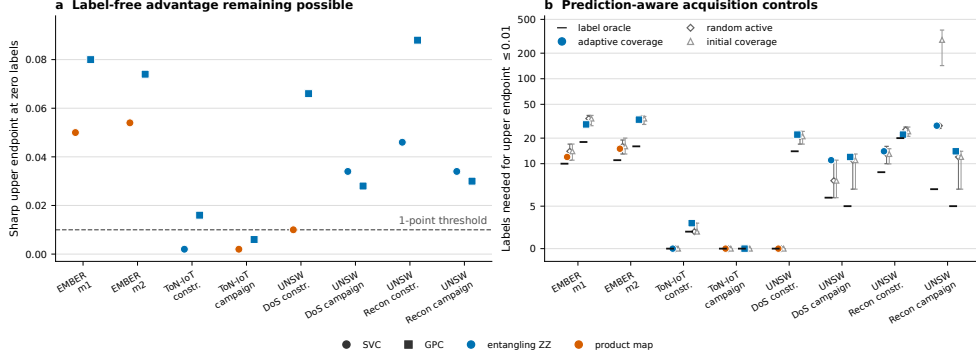
$$\mathcal{I}_B(L) = \left[ \min_j \frac{S_j(L) - R_j(L)}{n}, \min_j \frac{S_j(L) + R_j(L)}{n} \right].$$

*The upper endpoint is non-increasing as labels are audited and collapses to the realized family advantage when  $L$  contains the full batch.*

We use this result as an active falsification audit, not as a new model-selection rule. At each step the models remain fixed; one tested rule queries the point covering the largest number of witnesses currently tying at the smallest reducible upper bound (Methods). In this retrospective fixed-batch analysis, the rule reduces the 0.010 endpoint using 0–33 of 500 labels, with median 13 across the 16 case–classifier cells (Figure 2b). Stronger controls remove any claim that the coverage heuristic itself is unusually efficient: dynamic random sampling restricted to current actionable disagreements and a fixed initial-coverage ordering both have median 12.5 labels. Against random-active sampling, the adaptive rule wins, ties, and loses in 5, 5, and 6 cells; against initial coverage the counts are 6, 4, and 6. An exact ex-post label oracle needs median 6.5 labels, and the adaptive rule’s median excess is 4.5. The main acquisition result is therefore that prediction-aware restriction makes a small audit possible, not that one heuristic dominates. Prediction-independent ordering remains a weak diagnostic (median 245.5) and is reported only in Supplementary Information.

## 2.3 A reference-breadth–target-supervision evidence frontier

Nested classical reference families and partial labels expose two different costs of falsification. Increasing the classical family can only decrease the certificate’s upper



**Fig. 2** Sharp target-batch certificates against the full 115-candidate classical family. **a**, Zero-label upper endpoints for the SVC and GPC quantum models fixed by P1'. Colour separates selected entangling-ZZ and product-map models; the dashed line marks 0.010 accuracy. **b**, Labels needed to reduce the endpoint to at most 0.010 (symmetric-log scale). Horizontal ticks are the exact retrospective label oracle; filled markers are the fixed adaptive coverage rule; open diamonds and triangles are medians for dynamic random-active and non-adaptive initial-coverage controls, with bars spanning 2.5–97.5% across 200 SHA-256 tie orders. All certificates use the canonical q1000 realization (master, q-split, and model seeds 42) and a 500-point target batch. Comparator ranges are tie-order sensitivities, not confidence intervals.

**Table 1** Sharp accuracy certificates for the eight fixed target batches against the full 115-candidate classical family.  $U_0$  is the zero-label upper endpoint;  $n_{.01}$  is the fixed adaptive-coverage count for  $U \leq 0.010$ ;  $\Delta_{\text{full}}$  is the point-identified quantum advantage after all 500 labels are unlocked. The canonical run, quantum model, and every classical candidate were fixed before this audit; target labels are never used to retune or reselect them.

| Target case            | SVC   |           |                        | GPC   |           |                        |
|------------------------|-------|-----------|------------------------|-------|-----------|------------------------|
|                        | $U_0$ | $n_{.01}$ | $\Delta_{\text{full}}$ | $U_0$ | $n_{.01}$ | $\Delta_{\text{full}}$ |
| EMBER $m1$             | .050  | 12        | -.024                  | .080  | 29        | -.030                  |
| EMBER $m2$             | .054  | 15        | -.036                  | .074  | 33        | -.016                  |
| ToN-IoT constructed    | .002  | 0         | -.004                  | .016  | 3         | -.014                  |
| ToN-IoT campaign       | .002  | 0         | -.018                  | .006  | 0         | -.012                  |
| UNSW DoS constructed   | .010  | 0         | -.016                  | .066  | 22        | -.052                  |
| UNSW DoS campaign      | .034  | 11        | -.026                  | .028  | 12        | -.032                  |
| UNSW Recon constructed | .046  | 14        | -.032                  | .088  | 22        | -.066                  |
| UNSW Recon campaign    | .034  | 28        | -.026                  | .030  | 14        | +.002                  |

endpoint; auditing informative target labels can only decrease it further. We therefore report the surface  $(B, |L|) \mapsto U_B(L)$  and, for a materiality threshold  $\tau$ , the smallest reference breadth and target supervision required to establish  $U_B(L) \leq \tau$ . Here  $B$  indexes a prespecified *reference breadth*, not runtime or computational complexity.

The frozen block-ordering analysis already provides three nested breadths: 30, 60, and 115 candidates. For each cell we take the median zero-label endpoint over 5,000

507 prespecified whole-block orderings, then summarize the 16 case-classifier cells. The  
 508 cross-cell median contracts from 0.042 at  $B = 30$  to 0.034 at  $B = 60$  and remains  
 509 0.034 at  $B = 115$ ; the number of cells already at or below 0.010 changes from 3  
 510 to 4 to 4 (Supplementary reference-breadth table). Thus most aggregate contraction  
 511 occurs by 60 candidates, but the full family remains necessary for the deliberately  
 512 adversarial question and for individual cases. These ordering distributions quantify  
 513 dependence on reference breadth; they are not confidence intervals or computational-  
 514 cost measurements.

520     The prediction-aware evidence curves contract rapidly because only disagreements  
 521 capable of lowering a current witness endpoint need adjudication (corresponding curve  
 522 display in Supplementary Information). The adaptive rule is not consistently better  
 523 than random-active sampling, although recomputing the active set avoids the 286.5-  
 524 label median outlier produced by a frozen initial-coverage ordering. The result persists  
 525 when the selected quantum model uses an entangling-ZZ map: across the 12 entangling  
 526 case-classifier cells, the adaptive rule needs 0–33 labels (median 14) for the 0.010  
 527 endpoint, while random-active cell medians are 0–34 (median 12.5). Thus the finding  
 528 is not an artefact of including classically factorizable product maps.

535     Reference breadth matters separately from fair benchmarking. For ToN-IoT scan-  
 536 ning under GPC, the customary 30-candidate linear/RBF family yields a realized  
 537 quantum accuracy advantage of +0.028, so no number of labels can falsify a one-  
 538 point claim against that restricted reference. Against the full 115-candidate family the  
 539 realized advantage is  $-0.014$ , the zero-label upper endpoint is 0.016, and three tar-  
 540 geted labels reduce it below 0.010. Equal 60-versus-60 budgets remain the appropriate  
 541 performance comparison; the full family asks the different, deliberately adversarial  
 542 question of predictive indispensability.

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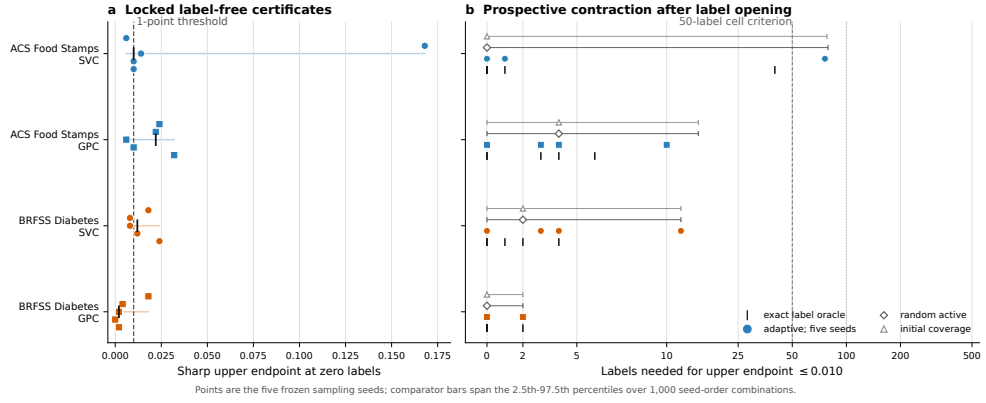
## 2.4 Prospective Gate-2 corroboration against the prespecified classical-kernel reference family

The security acquisition curves above are retrospective. We therefore specified a separate Gate-2 replication against the prespecified classical-kernel reference family before acquiring its three TableShift tasks, computing a model prediction, or opening a target label. The task identifiers were BRFSS Diabetes, ACS Food Stamps, and NHANES Lead. The immutable specification fixed five label-blind sampling seeds, the q1000 split sizes, P1' selection, both classifiers, the 60-quantum and 115-classical pools, physical prediction-label separation, three thresholds, 200 SHA-256 orders for each strong control, and numerical outcome criteria. The aggregate prediction lock was hashed before the one-time label audit (Methods).

All five NHANES seeds produced only 11 non-constant post-encoding features, below the frozen 12-dimensional grid requirement. The task therefore failed a pre-specified minimum-feature gate before any model candidate was executed; it was not replaced or repaired by reducing the grid. The replication is consequently technically limited to ten task-seed units, or 20 fixed task-seed-classifier cells, rather than evidence from three complete tasks.

The two technically eligible tasks met the predefined numerical criteria (Figure 3). Against the full prespecified classical-kernel reference family, zero-label endpoints range from 0 to 0.168 (median 0.010), and 11/20 cells already satisfy  $U \leq 0.010$  without target labels. The fixed adaptive audit reduces every remaining endpoint to that threshold with 0–76 labels (overall median 0). The four task-classifier medians are 0 for ACS SVC, 3 for ACS GPC, 3 for BRFSS SVC, and 0 for BRFSS GPC; all are below the 50-label criterion, and the worst seed remains below the predefined 100-label ceiling. Every realized accuracy effect is negative, ranging from  $-0.156$  to  $-0.002$ .

The prospective controls reproduce the earlier negative algorithmic conclusion. Against both dynamic random-active and non-adaptive initial-coverage medians,



**Fig. 3** Prospectively specified Gate-2 replication against the full 115-candidate prespecified classical-kernel reference family. **a**, Zero-label upper endpoints for five deterministic sampling seeds in each technically eligible task-classifier cell; black ticks mark medians and the dashed line marks 0.010 accuracy. **b**, Labels required to reduce the endpoint to at most 0.010. Coloured points are the five fixed adaptive counts, black ticks are exact retrospective label-oracle counts, and open markers with bars are medians and 2.5–97.5% quantiles for dynamic random-active and non-adaptive initial-coverage policies across 1,000 seed-order combinations (five seeds times 200 SHA-256 orders). The 50-label line is the predefined task-classifier median criterion; the dotted 100-label line is the predefined seed-level ceiling. Comparator ranges are algorithmic sensitivities, not confidence intervals. NHANES Lead failed the prespecified 12-feature gate before model execution and is absent.

adaptive coverage wins, ties, and loses in 3, 15, and 2 of the 20 cells, respectively; the median paired difference is zero. Prediction-independent ordering is far less efficient in the non-trivial cells. Thus the transferred result concerns the sharp certificate and restriction to actionable disagreements, not superiority of the coverage heuristic. All prospectively selected quantum winners are product maps, so this replication does not independently validate the entangling-ZZ stratum; that stratum remains supported only by the retrospective security cases.

## 2.5 The apparent advantage under oracle selection

We begin with a deliberately optimistic reference comparison: fidelity-based quantum kernels against linear and RBF baselines, with each family’s representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization

$C = 1$ . Related intrusion-detection studies motivate the small-reference context [34–36], but we do not claim that this exact protocol dominates QML practice. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 5 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing  $C = 1$  also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

## 2.6 The controlled comparison does not preserve the apparent advantage

A deployable comparison must tune each configuration’s regularization without looking at the test data, select the deployed configuration without OOD labels, and compare families at the same candidate budget. We implement all three (Section 4). For every kernel candidate, the SVC regularization  $C$  is chosen by five-fold cross-validation on the training Gram matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out from the ID test split (protocol P1’); no OOD label is ever consulted in selection. The primary comparison uses 60 candidates per extended family in every scenario-group and classifier, with the full 115-candidate classical pool retained as a sensitivity. We treat the eight scenario-groups as fixed case studies and report, per group, the quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population  $p$ -value (Section 4).

Table 2 and Figure 4 show no consistent evidence of a quantum advantage under the controlled endpoint. Against the equal-budget extended classical family,

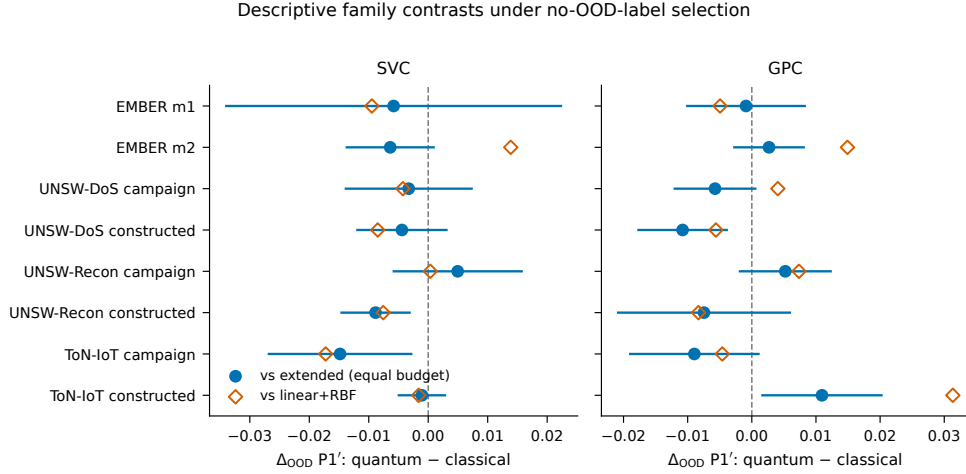
**Table 2** Target-label-free selection results for eight fixed scenario-groups. Values are quantum-minus-classical OOD balanced-accuracy differences under  $P1'$  (select on ID-validation, tune per-configuration  $C$  on train, evaluate on OOD test). The frozen extended-family analysis uses 60 candidates per family in every group-classifier cell. Each scenario interval is a 95% conditional cluster- $t$  interval over five q-split seed clusters ( $n = 5$ ), conditional on the benchmark pools and design; the fifteen pipeline realizations per setting are nested within these clusters. An interval containing zero is not an equivalence result. Source-dataset-equal rows average groups within three source datasets and report leave-one-source-dataset-out ranges, not confidence intervals. The customary linear+RBF comparison is contextual rather than equal-budget.

| Scenario                       | Clf. | vs lin./RBF | vs ext. (equal $B$ ) | interval / LODO range |
|--------------------------------|------|-------------|----------------------|-----------------------|
| EMBER $m1$                     | SVC  | -0.0095     | -0.0058              | $[-0.0340, +0.0223]$  |
|                                | GPC  | -0.0050     | -0.0009              | $[-0.0101, +0.0083]$  |
| EMBER $m2$                     | SVC  | +0.0139     | -0.0064              | $[-0.0137, +0.0009]$  |
|                                | GPC  | +0.0149     | +0.0027              | $[-0.0027, +0.0081]$  |
| UNSW-DoS (campaign)            | SVC  | -0.0043     | -0.0033              | $[-0.0139, +0.0073]$  |
|                                | GPC  | +0.0041     | -0.0057              | $[-0.0120, +0.0005]$  |
| UNSW-DoS (constructed)         | SVC  | -0.0085     | -0.0044              | $[-0.0119, +0.0031]$  |
|                                | GPC  | -0.0056     | -0.0108              | $[-0.0177, -0.0039]$  |
| UNSW-Recon (campaign)          | SVC  | +0.0004     | +0.0050              | $[-0.0058, +0.0157]$  |
|                                | GPC  | +0.0074     | +0.0052              | $[-0.0019, +0.0123]$  |
| UNSW-Recon (constructed)       | SVC  | -0.0076     | -0.0089              | $[-0.0146, -0.0031]$  |
|                                | GPC  | -0.0083     | -0.0075              | $[-0.0208, +0.0059]$  |
| ToN-IoT (campaign)             | SVC  | -0.0172     | -0.0148              | $[-0.0268, -0.0029]$  |
|                                | GPC  | -0.0046     | -0.0090              | $[-0.0190, +0.0010]$  |
| ToN-IoT (constructed)          | SVC  | -0.0016     | -0.0011              | $[-0.0049, +0.0028]$  |
|                                | GPC  | +0.0313     | +0.0109              | $[+0.0017, +0.0202]$  |
| Source-dataset-equal mean, SVC |      | -0.0041     | -0.0056              | $[-0.0070, -0.0045]$  |
| Source-dataset-equal mean, GPC |      | +0.0059     | -0.0009              | $[-0.0019, +0.0009]$  |

the source-dataset-equal difference is  $-0.00565$  for SVC (leave-one-source-dataset-out range  $[-0.00702, -0.00450]$ ) and  $-0.00094$  for GPC ( $[-0.00190, +0.00094]$ ). Against the customary linear+RBF reference, the corresponding differences are  $-0.00407$  and  $+0.00591$ , with heterogeneous group signs. UNSW-NB15 reconnaissance under held-out-campaign shift has a small positive mean ( $+0.005$  for both classifiers), with pointwise conditional intervals spanning zero. The full-pool extended-family sensitivity is similar in direction and scale ( $-0.00378$  for SVC and  $-0.00293$  for GPC).

Two secondary checks do not alter this fixed-case conclusion. Source-dataset-equal estimates remain close across blocked, uniform, and kernel-stratified equal-count sampling (SVC  $-0.00576$  to  $-0.00548$ ; GPC  $-0.00253$  to  $-0.00185$ ). Observational nearest-rank pairings at the prespecified 1.25 caliper retain 75.2% of pairs and have





**Fig. 4** Descriptive target-label-free family contrasts in eight fixed scenario-groups. P1' OOD differences (quantum minus classical) are shown for SVC (left) and GPC (right). Filled circles compare against the equal-budget extended classical family; error bars are pointwise 95% conditional cluster- $t$  intervals over five q-split clusters ( $n = 5$ ), conditional on the fixed benchmark pools and design. Open diamonds compare against the customary linear+RBF reference. The intervals are not simultaneous and do not imply population inference or equivalence.

negative median quantum-minus-classical differences in all eight groups; one-to-one and caliper sensitivities agree. Supplementary Tables 2–4 and Supplementary Figure 3 give the complete schemes, counts, and cautions; none identifies a causal family effect.

## 2.7 The quantum pool mixes entangling and product kernels

Circuit decomposition changes the interpretation of the aggregate family. The two ZZFeatureMap variants contain all-to-all two-qubit Pauli interactions, whereas the Pauli-X/Z and ZFeatureMap variants contain only one-qubit Pauli strings. For the latter pair, the prepared state and its fidelity kernel factorize over coordinates (Methods), permitting direct classical evaluation as a product of one-qubit overlaps. Thus two of the four map families cannot supply evidence that entanglement or hard quantum kernel estimation is responsible for their behaviour.

**Table 3** Logical circuit audit for the four quantum feature maps. Depth and CX columns give the values at 4 and 12 qubits after Qiskit 2.3.1 decomposition to **rz**, **sx**, **x**, and **cx**, optimization level 0, without device routing. “Fidelity CX” counts the compute–uncompute template for one kernel entry. Complete values at all five dimensions and one-qubit gate counts are provided in Supplementary Table 13 and the machine-readable artifact. These logical counts are not hardware-execution estimates.

| Map               | Pauli strings | Stratum           | Product factorization | map depth (4/12) | map CX (4/12) | fidelity CX (4/12) |
|-------------------|---------------|-------------------|-----------------------|------------------|---------------|--------------------|
| ZZ, 1 rep.        | Z, ZZ         | entangling ZZ     | no                    | 19/67            | 12/132        | 24/264             |
| ZZ, 2 reps.       | Z, ZZ         | entangling ZZ     | no                    | 35/107           | 24/264        | 48/528             |
| Pauli-X/Z, 1 rep. | X, Z          | separable product | yes                   | 11/11            | 0/0           | 0/0                |
| Z, 2 reps.        | Z             | separable product | yes                   | 8/8              | 0/0           | 0/0                |

Table 3 reports logical, unrouted resource counts. At 12 qubits, one ZZFeatureMap state preparation requires 132 CX gates for one repetition and 264 for two; the corresponding compute–uncompute fidelity templates require 264 and 528 CX gates. Both product-map templates require zero CX gates at every studied dimension. Under P1’ selection, product maps supply 594/1,080 (55.0%) of the selected quantum SVC configurations and 348/1,080 (32.2%) of the GP selections. The composition alone does not compare predictive quality; the matched stratum analysis below does so at 30 candidates per stratum.

At matched 30-candidate budgets, the source-dataset-equal SVC effect is  $-0.00928$  for the entangling-ZZ stratum and  $-0.00547$  for the product-map stratum; the constituent source-dataset means range from  $-0.01158$  to  $-0.00580$  and from  $-0.01167$  to  $+0.00021$ , respectively. The entangling-ZZ point estimate is negative in every fixed scenario-group ( $-0.01960$  to  $-0.00132$ ); product-map estimates range from  $-0.02245$  to  $+0.00571$ . The full 60-candidate pool remains at  $-0.00566$ . Thus the controlled result is not produced by pooling product maps with entangling circuits, and restricting the quantum family to ZZ interactions does not reveal a hidden positive family effect. Supplementary Table 8 reports every pointwise conditional interval; these strata are fixed-case sensitivities, not new population tests.

## 2.8 Evaluation choices interact within the controlled protocol

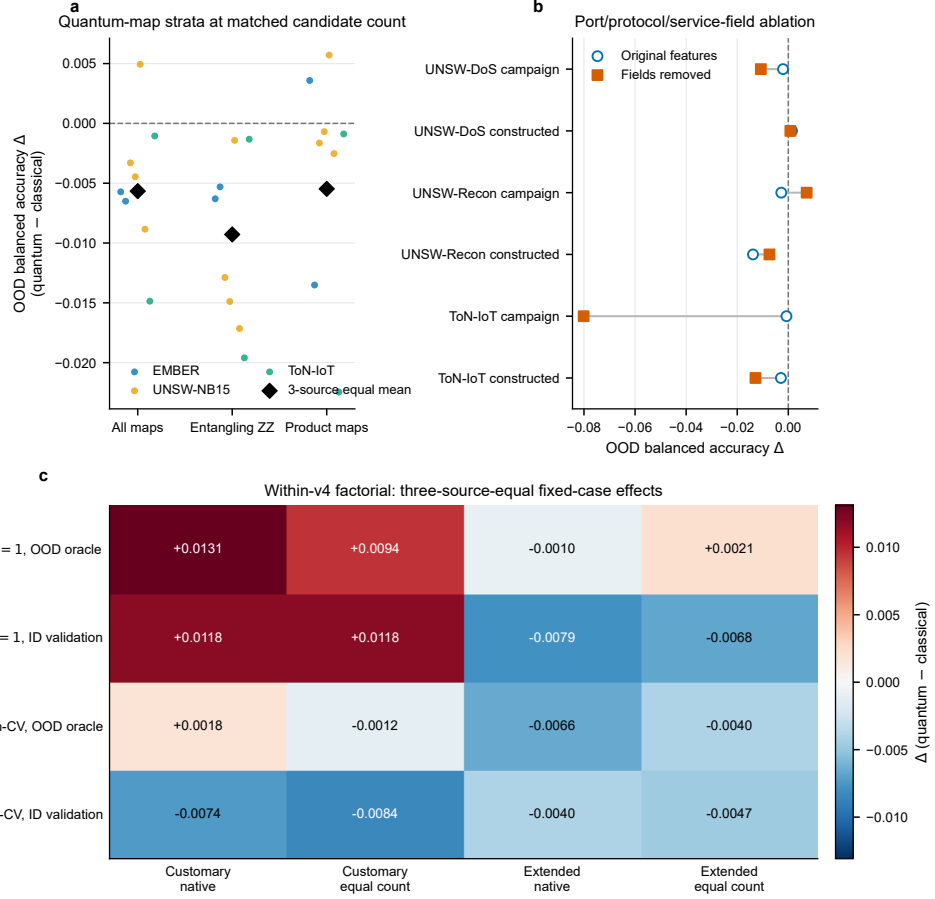
The evaluation-choice factorial holds the data generation fixed while varying four evaluation choices. In the q1000 SVC stratum, the fixed- $C$ , same-test-oracle, customary-reference, native-budget endpoint is  $+0.0131$ , whereas the train-CV, ID-validation, extended-reference, equal-count endpoint is  $-0.0047$ , a descriptive contraction of  $0.0178$ . The 16 three-source-dataset-equal endpoints range from  $-0.0084$  to  $+0.0131$  (Figure 5; Supplementary Table 10).

The largest paired mean changes arise from train-CV regularization ( $-0.0084$ ) and the extended reference ( $-0.0079$ ), but a regularization-by-reference interaction of  $+0.0139$  prevents additive or causal attribution (Supplementary Table 11). These controls therefore interact and must be reported together rather than attributed individually.

Removing the frozen port-, protocol-, and service-related fields also changes the relative comparison, confirming that common feature access alone does not make shortcut use family-invariant. Across the two network source datasets, the controlled effect moves from  $-0.0031$  to  $-0.0245$  (change  $-0.0215$ ; source-dataset changes  $-0.0447$  for ToN-IoT and  $+0.0018$  for UNSW-NB15). The six group changes range from  $-0.0794$  to  $+0.0099$ , and every pointwise change interval includes zero (Supplementary Table 12). The large negative change is concentrated in the ToN-IoT campaign shift, where removing direct traffic descriptors reduces both families' absolute accuracy but the selected fidelity kernel more strongly. This sensitivity neither restores a positive aggregate quantum effect nor establishes that all shortcuts have been removed.

## 2.9 Protocol sensitivity is prospectively corroborated on external domain shifts

We next applied the prospectively frozen protocol to College Scorecard, diabetes readmission, and ACS income, preserving the published TableShift domain splits and



**Fig. 5** Circuit-aware and evaluation-choice fixed-case sensitivities for SVC. **a**, Quantum-minus-classical OOD balanced-accuracy effects for all quantum maps ( $B = 60$ ), entangling-ZZ maps ( $B = 30$ ), and separable-product maps ( $B = 30$ ), each against a blocked equal-count extended-classical reference. Coloured points are the eight scenario-groups and black diamonds are three-source-dataset-equal means. **b**, Original and ablated controlled effects after removing the frozen port-, protocol-, and service-related fields in the six q1000 network groups. Connecting lines show paired group changes. **c**, Three-source-dataset-equal effects for the q1000 evaluation-choice  $2 \times 2 \times 2 \times 2$  factorial. Positive values favour the fidelity-kernel family. All quantities are descriptive summaries conditional on the fixed datasets and finite candidate pools; panels do not show population uncertainty or causal attribution.

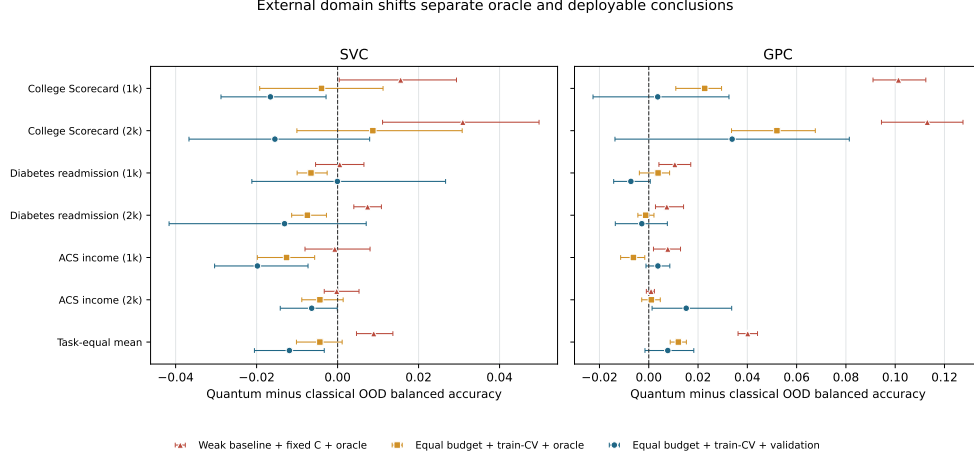
withholding all OOD labels from the controlled selector. The complete external grid contains 30 task-size-seed units and 37,800 split-level results; every prespecified unit passed the automated author-provided audit. Figure 6 shows all task-size effects and conditional seed-cluster intervals.

The primary SVC result satisfies the machine-readable outcome category frozen as `replicated_protocol_sensitivity`; narratively, protocol sensitivity is prospectively corroborated across three fixed external tasks. The task-equal controlled effect is  $-0.0119$  (95% conditional seed-cluster interval  $[-0.0205, -0.0033]$ ,  $n = 5$  deterministic subsampling seeds per task), whereas the weak-reference, fixed- $C$ , same-test-oracle effect is  $+0.0089$  ( $[+0.0047, +0.0137]$ ). Their contraction is  $+0.0208$  ( $[+0.0113, +0.0307]$ ), with a leave-one-task-out range of  $[+0.0116, +0.0260]$ . Size-equal controlled point estimates are  $-0.0160$  for College Scorecard,  $-0.0066$  for diabetes readmission, and  $-0.0131$  for ACS income; their pointwise intervals and both nested sizes are shown in Figure 6.

The secondary GP classifier shows a larger contraction of  $+0.0324$  ( $[+0.0223, +0.0427]$ ), but its task-equal controlled effect is  $+0.0077$  with an interval spanning zero ( $[-0.0015, +0.0183]$ ). College Scorecard leans quantum under this classifier, diabetes leans classical, and ACS is slightly positive. This classifier dependence limits the conclusion appropriately: the external study corroborates the sensitivity of the family verdict to evaluation design, not universal classical dominance.

## 2.10 Geometry is associated with robustness only partially

Effective rank and OOD kernel-target alignment are associated with OOD accuracy, but the association is regime-dependent. Across scenario-groups, median Spearman correlations span  $-0.15$  to  $+0.80$  for effective rank and  $+0.22$  to  $+0.64$  for OOD alignment; a leave-one-source-dataset-out ranking model reaches only  $0.22$ – $0.40$ . Because OOD alignment reuses the labels on which accuracy is measured, it is a post hoc diagnostic. Supplementary Figure 4 and Table 4 report the full analysis; neither descriptor is interpreted as a causal mediator or deployable selector.

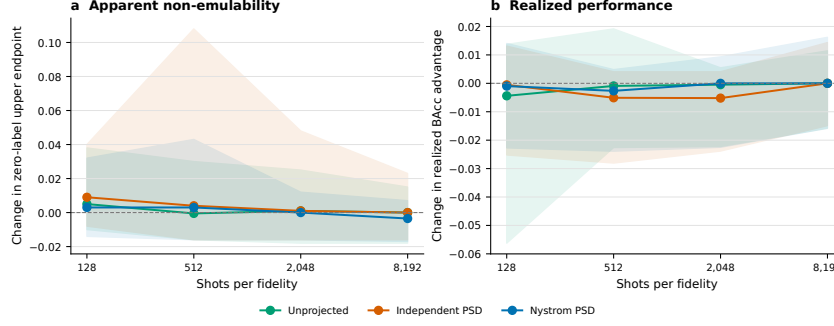


**Fig. 6** External domain shifts separate optimistic and deployable conclusions. Effects are quantum minus classical OOD balanced accuracy for the weak-reference fixed-regularization same-test oracle, the equal-budget train-tuned same-test oracle, and the primary equal-budget train-tuned validation selector. Rows show both nested sizes for all three fixed tasks plus the task-equal mean. Error bars are 95% conditional seed-cluster intervals over five deterministic subsampling seeds per task ( $n = 5$ ), preserving the nested-size relation. They quantify pipeline variation conditional on the fixed tasks and design, not uncertainty over a population of tasks.

## 2.11 Finite shots can create distinctness without useful advantage

All primary quantum kernels use exact statevectors. To test whether finite estimation can create apparent non-emulability, we apply the same zero-label certificate to 30 repeated binomial fidelity estimates at 128, 512, 2,048, and 8,192 shots for one fixed SVC quantum model per case. Each replicate reselects  $C$  from training data under unprojected, independently projected, and coherent Nyström-PSD conditions (Methods).

At 128 shots the perturbed model disagrees with its exact-statevector counterpart on a median 7.2–7.4% of target decisions, depending on PSD handling. Yet the across-case median increase in the full-family zero-label endpoint is only +0.003 for Nyström, +0.005 unprojected, and +0.009 for independent projection; individual cases range from decreases of 0.014 to increases of 0.040. The corresponding median



**Fig. 7** Finite-shot perturbations can increase apparent non-emulability without increasing performance. Lines are medians across the eight fixed case-level Monte Carlo medians and ribbons span their minima and maxima. **a**, Change from the exact-statevector zero-label accuracy upper endpoint against the full classical family. **b**, Change in realized OOD balanced-accuracy advantage. Each case has 30 measurement replicates at each shot level and PSD condition. Ribbons are fixed-case ranges, not confidence intervals.

realized balanced-accuracy changes are  $-0.0010$ ,  $-0.0044$ , and  $-0.0005$ , respectively (Figure 7). As shots increase, median certificate changes generally approach zero, while the case ranges remain heterogeneous.

Finite-shot noise can therefore move a quantum predictor away from every exact classical witness without making it better. This is a specifically important insufficiency result for the certificate: predictive non-emulability is necessary for a material advantage against the reference family, but it is not sufficient, and measurement noise can manufacture distinctness. Effective-rank inflation and the original performance sensitivities remain in the Supplementary Information. One case-replicate requires 1,624,250 distinct fidelity estimates; the full four-shot-level sensitivity projects 4.24 trillion circuit shots before routing and device overhead, halved if the four selected product-map cases are evaluated by direct factorization instead.

### 3 Discussion

The central result is an exact answer to a deliberately narrower question than quantum advantage in the complexity-theoretic sense: given fixed target inputs, fixed predictions, a bounded loss, and a prespecified classical-kernel reference family, how much

1059 predictive advantage remains compatible with the labels not yet inspected? The finite  
 1060 max–min gives the interval hull of the identified set after any partial audit without a  
 1061 sampling, calibration, prevalence, or shift assumption. No interval based only on those  
 1062 observables can be uniformly smaller. For zero–one accuracy, the disagreement matrix  
 1063 collapses that optimization to a closed form. This turns “a classical model looks sim-  
 1064 ilar” into a sharp and minimal finite-batch falsification statement with an explicit  
 1065 materiality threshold.

1070 This scope also separates the result from certifiable-surrogate generalization  
 1071 bounds, which control a target model’s population risk through a surrogate with a gen-  
 1072 eralization guarantee and unlabeled disagreement [37]. Our models remain fixed, the  
 1073 estimand is advantage against the best member of a declared family, and every unau-  
 1074 dited label completion remains admissible. The acquisition rule is therefore secondary:  
 1075 it chooses which ambiguity to remove, while the certificate is pathwise exact after  
 1076 any realized audit subset. In QML, this target-decision object complements Huang’s  
 1077 train-sample geometric sieve rather than replacing it [13].

1084 The empirical frontier is much sharper than the benchmark effect alone. Against  
 1085 115 classical kernels, at most 33 of 500 labels under the fixed adaptive rule are required  
 1086 to rule out an accuracy advantage above one point in every retrospective SVC and GPC  
 1087 case. Strong prediction-aware controls are at least as efficient on average, with median  
 1088 12.5 labels versus 13, whereas an exact retrospective oracle needs median 6.5. This  
 1089 negative algorithmic result matters: most of the gain comes from adjudicating action-  
 1090 able disagreements, not from a privileged acquisition heuristic. The same certificate  
 1091 contraction holds for the selected entangling-ZZ subset. In the prospectively specified  
 1092 replication, all four eligible task–classifier medians require at most three labels and  
 1093 the overall seed-level median is zero; the controls again show no acquisition-heuristic  
 1094 advantage. It does not establish universal classical emulation: it identifies the quantum  
 1095 model’s decisions relative to the prespecified classical-kernel reference family on these  
 1096



particular target covariates. Its value is that every remaining ambiguity is visible in the certificate rather than hidden inside an average performance comparison.

The three-gate interpretation prevents stronger conclusions from being conflated. Protocol validity asks whether the reported comparison could be deployed; predictive indispensability asks whether a witness in the prespecified classical-kernel reference family reproduces the fixed target decisions; quantum relevance asks whether any surviving behaviour depends on a non-factorizable circuit and remains meaningful after accounting for estimation cost. Passing all three gates creates a candidate for deeper advantage analysis, not a proof of speedup. Here Gate 1 reverses the optimistic sign, Gate 2 rules out a material batch advantage with limited auditing, and prospective corroboration in two technically eligible tasks meets the predefined criteria. Gate 3 finds no rescue from the retrospective entangling stratum while finite-shot noise sometimes increases distinctness without performance. Because all prospective winners are product maps, the prospective experiment corroborates Gate 2 but adds no independent evidence about Gate 3.

From the perspective of distribution-shift research, this reinforces that robustness conclusions are conditional on the protocol [9–11, 38]. Within the controlled protocol, moving from the fixed- $C$ , same-test-oracle, customary-reference, native-budget corner to the fully controlled corner changes the q1000 estimate from +0.0131 to  $-0.0047$ . Train-CV regularization and the extended reference have the largest mean paired changes, but their sizeable interaction precludes an additive or causal reading. The cross-protocol display cannot isolate regularization either, so the larger +0.037 endpoint obtained under oracle selection with fixed regularization is interpretable only as the outcome of a complete evaluation protocol, not of any single choice.

The external analysis provides prospective corroboration beyond the constructed security shifts. Under a protocol frozen before result inspection, three education, health, and socioeconomic tasks corroborate a positive protocol contraction and a

negative controlled SVC effect. The result is stable to omitting any one task, yet heterogeneous by task and classifier: the GP endpoint retains a small positive task-equal lean whose interval crosses zero. The strongest defensible conclusion is therefore about evaluation sensitivity, not the universal ordering of kernel families.

Length scale is decisive for both classical and quantum kernel behaviour [39, 40]. With symmetric scale freedom and tuned regularization, the families occupy overlapping effective-rank and alignment regimes. These quantities are kernel descriptors, not unique signatures of quantum provenance; OOD alignment additionally reuses target labels and remains strictly post hoc.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components [41], and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow [42]. There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of  $10^3$ – $4 \times 10^3$  samples at 4–12 qubits) sit far below the concentration regime. The repeated finite-shot sensitivity adds an empirical caveat these theoretical results anticipate: effective rank from sampled fidelities is inflated by a median factor of 1.93 at 128 shots and 1.13 at 8,192 shots (Section 2.11). Exact effective rank is therefore a statevector idealization even at these modest qubit counts.

From a practical perspective, no consistent fidelity-kernel advantage is observed over the tuned classical family, while overlap estimation introduces circuit and shot costs absent from direct classical-kernel evaluation. A shortened-length-scale Laplacian is therefore an important inexpensive reference, not a universally superior choice. More generally, comparisons should keep representation and learner fixed, tune all regularization on training data, match search budgets, select without target labels, include a

|                                                                                         |      |
|-----------------------------------------------------------------------------------------|------|
| non-constructed shift where possible, and report correlated benchmarks as fixed cases.  | 1197 |
| Geometry descriptors remain useful diagnostics, but not deployable selectors unless     | 1198 |
| validated without target-label access.                                                  | 1199 |
|                                                                                         | 1200 |
| Several limitations define the scope of our conclusions.                                | 1201 |
|                                                                                         | 1202 |
| Our shift operationalization covers three reproducible mechanisms—within-class          | 1203 |
| sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign    | 1204 |
| shift on network traffic (an unseen attack campaign plus a capture-partition change,    | 1205 |
| not a timestamped temporal split)—but not the full space of real-world drift processes. | 1206 |
| Explicit temporal splits on timestamped malware, family-based drift, and adversarially  | 1207 |
| induced shift remain outside the study’s scope, and our conclusions should be re-tested | 1208 |
| under them rather than assumed.                                                         | 1209 |
|                                                                                         | 1210 |
| Kernel and encoding coverage also remains finite. The classical family spans seven      | 1211 |
| kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom,     | 1212 |
| but neither candidate set is exhaustive: the quantum side is restricted to four com-    | 1213 |
| monly used fidelity feature maps at low qubit counts, and trained or task-adapted       | 1214 |
| encodings, projected kernels, and deeper maps are not evaluated. The observed asso-     | 1215 |
| ciation suggests that future extensions should report their Gram-matrix geometry as     | 1216 |
| well as provenance, but it does not predict their outcome. At substantially larger      | 1217 |
| qubit counts, exponential kernel concentration [42] creates an additional regime not    | 1218 |
| represented here, so our conclusions should not be extrapolated beyond the low-qubit    | 1219 |
| study.                                                                                  | 1220 |
|                                                                                         | 1221 |
| The certificate is relative to a finite, prespecified classical family and a fixed      | 1222 |
| observed target batch. It does not rule out a quantum advantage against a differ-       | 1223 |
| ent classical family, establish a surrogate outside the observed covariates, or convert | 1224 |
| candidate count into computational complexity. Likewise, the batch interval is not      | 1225 |
| a confidence interval for a future deployment population. The optional population       | 1226 |
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1243 result requires i.i.d. target sampling and models fixed before observing that sample,  
1244 assumptions that do not describe the constructed tail batches.

1246     The partial-label result is mathematically valid for any audit subset, but the secu-  
1247 rity acquisition curve is retrospective. In the separate prospective Gate-2 experiment  
1248 against the prespecified classical-kernel reference family, tasks, prediction locks, poli-  
1249 cies, thresholds, and numerical criteria were fixed and hashed before label opening. The  
1250 two technically eligible tasks met those predefined criteria. Its scope remains narrow:  
1251 NHANES failed a prespecified feature gate, the seeds are deterministic subsamples  
1252 rather than independent task replicates, and every prospective quantum winner is  
1253 a product map. The result corroborates the certificate and actionable-disagreement  
1254 principle on these fixed tasks, not a population claim, entangling-kernel replication,  
1255 or superiority of bottleneck coverage. Balanced-accuracy certification additionally  
1256 requires a prespecified prevalence or prevalence range; using the retrospectively  
1257 observed prevalence is only a post-unlock sensitivity.

1266     The quantum kernels are simulated rather than executed on hardware. The study  
1267 therefore characterizes fidelity-kernel geometry, not noisy-device behaviour. Repeated  
1268 binomial sampling quantifies one component of measurement uncertainty, but hard-  
1269 ware noise, transpilation, readout, mitigation, and circuit-dependent correlations  
1270 remain unmodelled [43]. Finite-shot cost is not merely an implementation detail: Gen-  
1271 tinetta et al. place kernel-estimation uncertainty inside the end-to-end complexity  
1272 of quantum support-vector machines [44], while exponential concentration can make  
1273 informative spectral structure expensive to resolve at larger scales [42]. Adaptive allo-  
1274 cation of a limited measurement budget across kernel entries has been reported to  
1275 improve classifier fidelity and measurement efficiency relative to uniform allocation  
1276 [45]; our repeated-shot sensitivity uses uniform allocation and does not optimize it.  
1277 Recent QML evidence is not uniformly negative. Agliardi et al. report a favorable  
1278 10-qubit hardware instance on real data (87% accuracy versus 80% for their classical  
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benchmark) and comparable utility-scale results on 156-feature synthetic data (80% versus 83%) [46]. Those study-specific, single-instance comparisons motivate continued hardware evaluation but do not alter our fixed-batch result or validate advantage under our shift protocol. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [47]. At the dimensions studied, the fidelity kernels are also classically simulable [14, 48], and the ZZ map admits a closed-form classical surrogate in the small-bandwidth regime [15]; this is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats provide a final boundary. Dynamic malware analysis and encrypted traffic remain untested. The network exports retain port-derived counters and, for ToN-IoT, direct port, protocol, and service indicators that can support shortcuts in NIDS benchmarks [49]. Sharing features controls the inputs but does not imply that distinct kernels exploit them equally. In the frozen q1000 ablation, removing these fields shifts the two-source-dataset-equal effect from  $-0.0031$  to  $-0.0245$ , but the change is highly heterogeneous and concentrated in ToN-IoT. This is evidence that feature access can affect the relative comparison, not proof that every shortcut has been removed or that the ablated tasks define a preferred deployment benchmark.

We do not claim universal classical superiority, predictive equivalence, complexity-theoretic advantage, hardware relevance, or a refutation of quantum kernels. The proposed audit instead makes a candidate advantage progressively falsifiable: first by protocol controls, then by a prespecified classical witness family on target covariates, and finally by limited adjudication of the disagreements that keep a material advantage possible. Future work should broaden the prospective replication to more

tasks and entangling winners, and pair candidate breadth with measured computational resources. This interpretation is consistent with calls for controlled QML benchmarking [4, 6, 50].

## 4 Methods

### 4.1 Problem setting and estimands

We study binary security classification—malware versus benign software and attack versus benign traffic—under distribution shift. Let  $D_{\text{ID}}$  denote the in-distribution process and  $D_{\text{OOD}}$  a constructed stress-test or held-out-campaign process. Each run contains a training set  $T$ , an ID holdout that is divided into validation  $V_{\text{ID}}$  and test  $S_{\text{ID}}$ , and an OOD test set  $S_{\text{OOD}}$ . Preprocessing, samples, classifier implementation, tuning rule, and family-selection rule are shared; only the kernel changes. Comparisons are always within a classifier.

The primary endpoint is OOD balanced accuracy after P1' deployment selection. A family effect is

$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)},$$

so positive values favour the quantum family. ID balanced accuracy and ID-to-OOD degradation are secondary. P2 selects from OOD information in other realizations, and P3 selects and evaluates on the same OOD test labels; both are explicitly non-deployable descriptive oracles.

### 4.2 Sharp finite-target identification

For Theorem 1, fix the prediction-dependent loss tables before the audit and write  $a_{ij}(y) = \ell_{ij}^C(y) - \ell_i^Q(y)$ . For any completion of the labels outside  $L$ ,

$$\Delta_\ell(y) = \frac{1}{n} \min_j \left\{ S_j(L) + \sum_{i \notin L} a_{ij}(y_i) \right\}.$$

For the lower endpoint, minima commute and the label choices separate by item:

$$\min_{y \in L^c} \min_j \left\{ S_j + \sum_{i \notin L} a_{ij}(y_i) \right\} = \min_j \left\{ S_j + \sum_{i \notin L} \min_y a_{ij}(y) \right\}.$$

For the upper endpoint introduce one-hot variables  $z_{iy} \in \{0, 1\}$  with  $\sum_{y \in \mathcal{Y}} z_{iy} = 1$  and maximize  $t$  subject to

$$nt \leq S_j(L) + \sum_{i \notin L} \sum_{y \in \mathcal{Y}} a_{ij}(y) z_{iy} \quad \text{for every } j.$$

This mixed-integer linear program enumerates the finite max–min implicitly. Finiteness of  $\mathcal{Y}^{|L^c|}$  guarantees that both extrema are attained. Any interval based on the same loss tables and audited labels that excluded either endpoint would fail on the attaining completion, proving sharpness and minimality. Revealing a label restricts the completion set, which proves monotone contraction. The implementation is tested against exhaustive enumeration for binary Brier loss, arbitrary multiclass bounded-loss tables, arbitrary partial labels, and the zero–one specialization.

The empirical accuracy certificate changes the estimand from a selected family mean to a fixed-model comparison. Let  $q_i = h_Q(x_i)$  and  $c_{ij} = h_{C_j}(x_i)$  be hard predictions on a target batch of size  $n$ . The quantum model is the P1' winner fixed using ID validation; the classical candidates, their train-only hyperparameters, and their target predictions are also fixed before any target-label audit. Every displayed certificate uses the same inherited canonical realization per security group:  $n_{\text{train}} = 1000$ ,  $n_{\text{ID}} = 500$ ,  $n_{\text{OOD}} = 500$ , master seed 42, q-split seed 42, and model seed 42. The exact scenario run prefixes and selected SVC/GPC configurations appear in the canonical-run table in Supplementary Information. For a candidate labelling  $y$ ,

$$g_j(y) = \text{Acc}(q; y) - \text{Acc}(c_j; y), \quad \Delta_B(y) = \min_j g_j(y).$$

1427 Only disagreements contribute to  $g_j$ . If  $d_j = n^{-1} \sum_i \mathbf{1}\{q_i \neq c_{ij}\}$ , then  $g_j(y) \in$   
1428  $[-d_j, d_j]$ . The labelling  $y_i = q_i$  attains  $g_j = d_j$  for every  $j$  simultaneously, so  
1429  $\max_y \Delta_B(y) = \min_j d_j$ . If  $k$  maximizes  $d_j$ , the labelling  $y_i = c_{ik}$  attains  $\Delta_B(y) \leq$   
1430  $g_k(y) = -d_k$ ; no  $g_j$  can be below  $-d_j \geq -d_k$ , so  $\min_y \Delta_B(y) = -\max_j d_j$ . This proves  
1431 Theorem 2 and its sharpness.

1435 For an arbitrary audited subset  $L$ , define the observed signed contribution  $S_j(L)$   
1436 and remaining disagreement count  $R_j(L)$  as in Theorem 3. Every completion satisfies  
1437

$$\frac{S_j - R_j}{n} \leq g_j(y) \leq \frac{S_j + R_j}{n}.$$

1442  
1443 Assigning every unaudited label to  $q_i$  attains all witness upper endpoints simultane-  
1444 ously. Assigning them to the witness that minimizes  $S_j - R_j$  attains the lower endpoint  
1445 of  $\Delta_B$ . This proves Theorem 3. In binary classification, if  $D_j$  is the original disagree-  
1446 ment count and the audit has revealed  $a_j$  disagreements favouring  $q$  and  $b_j$  favouring  
1447  $c_j$ , the interval simplifies to

$$\left[ \min_j \frac{-D_j + 2a_j}{n}, \min_j \frac{D_j - 2b_j}{n} \right].$$

1457 Thus the upper endpoint cannot increase under arbitrary adaptive querying. A  
1458 classical-favouring label on a binary disagreement reduces that witness endpoint by  
1459 exactly  $2/n$ ; a quantum-favouring label leaves its upper endpoint unchanged. These  
1460 statements are deterministic for the observed batch and require neither i.i.d. sampling  
1461 nor a model for the unaudited labels.

1465 For completeness, an optional population form is available when the target inputs  
1466 are i.i.d. and every model is fixed before observing them. A union bound and



Hoeffding’s inequality give, with probability at least  $1 - \delta$ ,

$$\sup_y \Delta_B^{P_T}(y) \leq \min_{j \leq M} \hat{d}_j + \sqrt{\frac{\log(2M/\delta)}{2n}}.$$

We do not apply this result to the constructed tail batches, for which the finite-batch certificate is the appropriate estimand.

### 4.3 Reference tiers and active target-label audit

The partial-label audits use two nested operational tiers. The customary tier contains the 30 linear and RBF candidates; the full tier contains all 115 prespecified classical candidates. Separately, the frozen zero-label breadth sensitivity evaluates 30, 60, and 115 candidates using 5,000 nested whole-block orderings rooted at `ksf-v9-frontier-20260731`; a 60-candidate draw contains 12 of 23 five-dimension blocks and the full tier contains all blocks. Tier size measures reference breadth only. It is not a computational-resource claim because training time, memory, and model-construction cost are not equalized by candidate count.

At every audit step, we calculate the witness-specific numerator  $S_j + R_j$ . Among witnesses with unaudited disagreements, the adaptive coverage rule identifies those tying at the smallest reducible numerator and queries an unaudited input that covers the largest number of these bottleneck witnesses. Ties follow a fixed SHA-256 order. The rule uses target covariates, predictions, and previously revealed outcomes, but never an unrevealed label. It is evaluated retrospectively for thresholds  $\tau \in \{0.005, 0.010, 0.020\}$ .

We compare it with three stronger references. Dynamic random-active sampling selects by SHA-256 order among all unaudited points disagreeing with at least one current reducible bottleneck witness. Non-adaptive initial coverage ranks points once by how many zero-label bottleneck witnesses they cover, without label feedback. Both

1519 use 200 frozen SHA-256 tie orders. Finally, an undeployable label oracle gives the exact  
1520 ex-post minimum number of queries over every possible audit subset. If  $D_j$  is witness  
1521  $j$ 's total disagreement count, crossing threshold  $\tau$  requires

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$$r_j(\tau) = \max\left\{0, \left\lceil \frac{D_j - n\tau}{2} \right\rceil\right\}$$

revealed disagreements whose realized label favours witness  $j$ ; the oracle minimum is the smallest feasible  $r_j(\tau)$  over witnesses. This is exact because the family endpoint crosses when any one witness endpoint crosses. A prediction-independent complete ordering is retained only as a weak diagnostic. Comparator quantiles describe frozen algorithmic orders, not sampling uncertainty.

The zero-label SVC definition and go/no-go criterion were frozen before prediction outcomes were inspected. The arbitrary-partial-label acquisition extension was declared only after the target labels had been unlocked; the strong-comparator amendment was declared after the original acquisition curves were known. Both are exploratory on these eight cases. GPC repeats the mathematics and policies but is not an independent prospective validation. Target labels are used only to evaluate certificate contraction, the oracle, and the final realized effect; no kernel, classifier, regularization, threshold, or P1' winner is refitted or reselected. Prediction archives and the separately stored label file are joined only after SHA-256, row-index, model, and target-length gates pass.

#### 1555 **4.4 Prospectively specified Gate-2 replication against the** 1556 **prespecified classical-kernel reference family**

1559 The Gate-2 replication against the prespecified classical-kernel reference fam-  
1560 ily was specified and hashed on 1 August 2026 before any of its three tasks  
1561 was acquired, any model prediction was computed, or any target label was

inspected. The source is the same TableShift implementation pinned at commit `fca9429814703a07e3902d005d46563a207b7f0a` [51]. The task identifiers were BRFSS Diabetes (race shift), ACS Food Stamps (Census-division shift), and NHANES Lead (poverty shift). All three were attempted and none could be replaced based on availability or outcome.

For each task, seeds  $\{42, 123, 999, 7, 2024\}$  define deterministic label-blind samples with  $(n_{\text{train}}, n_{\text{val}}, n_{\text{ID}}, n_{\text{target}}) = (1000, 250, 250, 500)$ . The v5 train-only semantic encoding, scaling, TruncatedSVD dimensions  $\{4, 6, 8, 10, 12\}$ , and angular representation are reused verbatim. The 60 quantum candidates and 115 classical candidates, P1' validation selection, candidate-specific train-CV for SVC, fixed Laplace GPC, and decision thresholds are unchanged. The customary 30-candidate tier is secondary; the full tier and  $\tau = 0.010$  define the prospective outcome. Secondary thresholds are 0.005 and 0.020.

The pipeline is fail-closed. A prediction-lock stage copies target labels to a separate sealed tree, removes them from the model state, and writes hard target predictions, row-order digests, selected validation-only quantum configurations, and SHA-256 hashes without calculating or printing target performance, prevalence, disagreement, or an acquisition curve. Candidate checkpoints contain only target predictions and validation-selection quantities. An aggregate manifest is accepted only when every available task-seed unit has all 175 kernel configurations for both classifiers or a permitted pre-model technical-failure record. The audit stage verifies that aggregate hash, records the one-time label opening and analysis-code hash, then joins labels by source-position digest. No model is refitted after the join.

For each classifier and the two classical tiers, the fixed policies are adaptive bottle-neck coverage, 200 dynamic random-active SHA-256 orders, 200 non-adaptive initial-coverage orders, the exact ex-post label oracle, and a weak prediction-independent order. The hash root appends task, seed, model, tier, policy, and draw. The primary

summaries are four or six task–classifier medians over five seeds, depending on technical availability. The predefined primary criterion requires every available task–classifier median to be at most 50 labels, the overall seed-level median to be at most 25, and no seed to exceed 100. With exactly two available tasks, all four cells are required and the result is explicitly technically limited. A secondary criterion requires all four medians to be at most 50 and an overall median at most 50; fewer than two available tasks is technically incomplete. The criteria were evaluated exactly once.

NHANES generated 11 non-constant encoded features in all five seeds, fewer than required for the frozen 12-dimensional candidate grid. Minimum-feature failure was an explicitly permitted technical-unavailability gate. No NHANES model candidate was executed, the task was not replaced, and dimensions were not changed. BRFSS Diabetes and ACS Food Stamps passed every gate, leaving ten task–seed units for the technically limited prospective outcome.

#### 4.5 Balanced-accuracy identification

Balanced accuracy is not label-free unless target prevalence is known or restricted. At a fixed positive count  $n_+$ , we group inputs by their joint prediction signature  $z = (q_i, c_{i1}, \dots, c_{iM})$ . Let  $n_z$  be the size of group  $z$  and  $k_z \in \{0, \dots, n_z\}$  its unknown number of positive labels, with  $\sum_z k_z = n_+$ . For classifier  $h$  with group-level prediction  $h_z \in \{0, 1\}$ ,

$$\text{BAcc}_h(k) = \frac{1}{2} \left[ \frac{\sum_z k_z h_z}{n_+} + \frac{\sum_z (n_z - k_z)(1 - h_z)}{n - n_+} \right].$$

The sharp upper endpoint is the mixed-integer linear program that maximizes  $t$  subject to

$$t \leq \text{BAcc}_Q(k) - \text{BAcc}_{C_j}(k) \quad \text{for every } j,$$

together with the integer and prevalence constraints above. The lower endpoint is the minimum over witnesses of their corresponding linear minima. Relaxing  $k_z$  to a continuous interval gives the sharp distributional or fractional-label LP. When prevalence

is known only to lie in an interval, we maximize over its admissible integer counts. We report this as a sensitivity because the natural target prevalence is unavailable without labels; accuracy remains the main assumption-free certificate.

#### 4.6 Why train-only geometry cannot determine target prediction

The geometric difference and effective rank summarize training Gram structure, whereas deployment also depends on the target–train block. An explicit normalized PSD construction makes the distinction exact. Set  $K_{TT} = I_n$ . One valid joint Gram extension gives a unit-norm target feature orthogonal to all training features, hence  $K_{UT} = 0$ . A second gives the target the same feature as the first training point, hence  $K_{UT} = e_1^\top$ . Both joint matrices are PSD, have unit diagonal, and share identical  $K_{TT}$ . For kernel ridge regularization  $\lambda > 0$ , however, their target prediction operators are

$$0 \quad \text{and} \quad e_1^\top (I_n + n\lambda I_n)^{-1} = \frac{e_1^\top}{1 + n\lambda},$$

whose spectral-norm difference is  $1/(1 + n\lambda)$ . Train-only geometry can therefore screen possible separation but cannot characterize target transport. We use the hard-prediction certificate for SVC and GPC rather than treating this KRR construction as an endpoint result.

#### 4.7 Security datasets and shared split sizes

The source datasets are EMBER [52], UNSW-NB15 [53], and ToN-IoT [54]. Four binary scenarios—EMBER malware, UNSW-NB15 DoS, UNSW-NB15 Reconnaissance, and ToN-IoT Scanning—produce two shift mechanisms each and hence eight scenario-groups. Classical and quantum kernels always receive identical row indices and class balance.

1703 The 72 principal settings cross eight groups with master seeds  $\{42, 123, 999\}$  and  
1704  $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$  equal to  $(1000, 500, 500)$ ,  $(2000, 1000, 1000)$ , or  $(4000, 1800, 1800)$ .  
1705 Each setting is instantiated with q-split seeds  $\{42, 123, 999, 7, 2024\}$  and model seeds  
1706  $\{42, 123, 999\}$ , yielding fifteen pipeline realizations arranged in five q-split clusters.  
1707 Indices are stored explicitly; automated gates verify split-role disjointness, required  
1708 cardinalities, both classes, identical family inputs, and absence of label-derived  
1709 features.  
1710

#### 1715 4.8 EMBER shift construction

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1717 EMBER uses the histogram and byte-entropy feature blocks and two label-conditional  
1718 tail stress tests. They are not described as certified covariate shifts because condition-  
1719 ing the split on class does not establish invariance of  $P(Y | X)$ . For  $m1$ , the sparsity  
1720 score is the number of non-zero entries in the selected feature block. Within each class,  
1721 low- and high-score extremes supply the exact-size OOD set; training and ID samples  
1722 are stratified draws from the remainder.

1723 For  $m2$ , the final training set is fixed first. A MaxAbs scaler and TruncatedSVD  
1724 representation are fitted on training data only, and a centroid is computed for each  
1725 class. Every non-training sample is scored by Euclidean distance to its own-class cen-  
1726 troid. Within-class low- and high-distance extremes form  $S_{\text{OOD}}$ , and the ID holdout is  
1727 sampled from the remaining pool. This construction probes unusual geometry relative  
1728 to the fixed training set without using test information to fit the representation.

#### 1738 4.9 Network-flow shift construction

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1740 Numeric preparation removes identifiers and explicit label derivatives, leaving 39  
1741 UNSW-NB15 and 88 ToN-IoT features. The UNSW-NB15 export excludes direct cat-  
1742 egorical protocol, service, and state fields but retains three port-derived counters;  
1743 ToN-IoT retains two port fields plus protocol and service indicators. A leakage audit  
1744 records every exclusion and feature-label association; the largest observed absolute

correlation is 0.62 and belongs to a retained traffic feature. Each scenario has reference  
and current pools and uses two mechanisms.

The *m2-centroid* mechanism is the network analogue of EMBER *m2*: train-only  
MaxAbs+TruncatedSVD class centroids define within-class distance tails, which sup-  
ply the OOD set. The *attack-campaign* mechanism performs no geometric selection.  
Training and ID pools contain benign traffic and all non-target attack categories;  
OOD contains benign traffic and the held-out target campaign. UNSW-NB15 refer-  
ence and OOD pools additionally use different official capture partitions. This is an  
unseen-campaign plus capture-partition shift, not a timestamped temporal split.

#### 4.10 Shared preprocessing and representations

For every run and dimension  $d \in \{4, 6, 8, 10, 12\}$ , a MaxAbs scaler and TruncatedSVD  
are fitted on  $T$ ; the reduced coordinates are standardized and mapped linearly with  
training-fitted parameters so that the *training* coordinates lie in  $[0, \pi]$ . Held-out coor-  
dinates receive the same affine map and may extrapolate outside that interval under  
shift. The same embedded samples are supplied to both families, making the compar-  
ison a kernel swap rather than a representation swap. Dimension is a candidate axis.  
No validation, ID-test, or OOD-test row fits a transformation. The SVD random state  
equals the model seed.

#### 4.11 Classifiers and train-only regularization

The primary learner is an SVC with a precomputed Gram matrix and balanced class  
weights [55]. For every kernel geometry,  $C \in \{0.01, 0.1, 1, 10, 100\}$  is selected by five-  
fold stratified cross-validation on the training Gram matrix; ties favour the smaller  $C$ .  
When  $n_{\text{train}} > 2000$ , this selection step uses a class-stratified random-state-20260717  
subsample capped at 2,000 rows, while the final model is always refitted on the com-  
plete training block. Because  $C$  is resolved before family selection, its grid points are  
not counted as separate family candidates.

The secondary learner is a binary Gaussian-process classifier with a logistic link and Laplace approximation [56]. It consumes the same precomputed kernel, uses diagonal jitter  $10^{-8}$ , at most 100 Newton iterations, and convergence tolerance  $10^{-6}$ . The logistic–probit approximation supplies predictive probabilities; no post-hoc calibration is applied.

#### 4.12 Classical kernel family

The customary reference contains a cosine-normalized linear kernel and RBF kernel. The extended family adds cosine-normalized degree-2 and degree-3 polynomial kernels with  $\gamma = 1/d$  and intercept 1, the Laplacian kernel  $\exp(-\|x-x'\|_1/\ell_1)$ , and Matérn-3/2 and Matérn-5/2 kernels. The base  $\ell_1$  and  $\ell_2$  scales are median pairwise Manhattan and Euclidean distances, respectively, calculated from at most 2,000 training rows with the model seed; the RBF base scale is fitted from training variance. RBF, Laplacian, and both Matérn kernels use factors  $\{0.1, 0.3, 1, 3, 10\}$  around these training-only bases. Deduplication yields 23 kernel-shape/scale blocks crossed with five dimensions, or 115 extended-classical candidates in full-coverage groups.

#### 4.13 Quantum fidelity-kernel family

The quantum family uses exact statevector fidelities

$$K_Q(x, x') = |\langle \phi_Q(x) | \phi_Q(x') \rangle|^2$$

[1, 2]. Qiskit 2.3.1 implements four map blocks [57]: fully connected ZZ maps with one and two repetitions; a one-repetition Pauli map with single-qubit strings X and Z; and a two-repetition Z map. Only the ZZ maps contain two-qubit interactions. The latter



two prepare product states,

$$U(x) = \bigotimes_{j=1}^d U_j(x_j), \quad K_Q(x, x') = \prod_{j=1}^d |\langle \phi_j(x_j) | \phi_j(x'_j) \rangle|^2,$$

so the constructor argument `entanglement="full"` stored for the Pauli-X/Z map has no effect without a multiqubit Pauli string. The product can be evaluated classically in  $O(d)$  one-qubit-overlap calculations. We call these the entangling-ZZ and separable-product strata. Each map uses angle scales  $\{0.5, 1, 2\}$  and all five dimensions, giving 60 candidates (30 per stratum) at 4–12 simulated qubits. Classical bandwidth and quantum angle-scale freedom both control locality [39, 40]; no circuit is trained.

#### 4.14 Circuit and sampling-resource audit

Circuit counts use Qiskit 2.3.1 and the logical basis  $\{\mathbf{rz}, \mathbf{sx}, \mathbf{x}, \mathbf{cx}\}$  at transpiler optimization level 0, without backend layout or routing. We report each feature-map template and the compute–uncompute fidelity template formed by composing the map at  $x$  with the inverse map at  $x'$  after assigning distinct numeric inputs. Counts are compiler- and basis-conditional logical resources, not device schedules.

For the repeated q1000 finite-shot sensitivity, each case and measurement replicate estimates the off-diagonal training triangle, the combined 500-row ID-holdout-to-training rectangle (partitioned downstream into validation and test roles), the OOD-test-to-training rectangle, and the off-diagonal OOD square triangle; diagonals are fixed to one. Thus one kernel configuration requires

$$\binom{1000}{2} + 500(1000) + 500(1000) + \binom{500}{2} = 1,624,250$$

distinct fidelity estimates per shot level and replicate. Multiplying this count by shots gives the idealized circuit-shot total. The three PSD conditions reuse the same sampled entries and add only classical post-processing. Routing, calibration, mitigation, queueing, and device noise are excluded.

#### 4.15 Target-label-free family selection

The original ID holdout is divided deterministically with class stratification into  $V_{ID}$  and  $S_{ID}$ . Within each class, global row indices are ordered by SHA-256 of the salt `ksf-v4-idsplit::` and index, then assigned alternately to validation and test. The halves are disjoint and approximately class-balanced. P1' deploys, within each family, the candidate with highest  $V_{ID}$  balanced accuracy and reports it once on  $S_{ID}$  and  $S_{OOD}$ . Ties follow lexicographic candidate order. OOD labels never enter regularization, preprocessing, or P1' selection.

#### 4.16 Candidate-budget matching

Best-of-family selection rewards a larger pool. The confirmatory coverage audit found 115 extended-classical and 60 quantum candidates in every one of the eight scenario-groups and both classifiers. The primary comparison therefore draws 60 candidates from each family in all 16 group-classifier cells. For each cell, 5,000 extended-classical subsamples are generated with seed 20260715. The primary `kernel.blocked` scheme samples 12 of the 23 complete kernel-shape/scale blocks, each crossed with all five dimensions, to mirror the 12 quantum feature-map/angle-scale blocks crossed with the same dimensions. This matches both candidate count and block structure, although it does not make the two search spaces identical. `uniform` samples individual candidates, while `kernel.stratified` distributes the draw across kernel-shape/scale strata. Each subsample selects its P1' winner; the classical endpoint is the resampling expectation. The full 115-candidate pool is secondary. Resampling percentiles describe sensitivity to the sampled candidate set, not population uncertainty.

|                                                                                                                               |      |
|-------------------------------------------------------------------------------------------------------------------------------|------|
| <b>4.17 Entangling and separable quantum strata</b>                                                                           | 1933 |
|                                                                                                                               | 1934 |
| The circuit-stratum sensitivity reuses all controlled-protocol summaries and compares                                         | 1935 |
| three SVC pools under train-CV and P1': the full 60-candidate quantum family, the 30                                          | 1936 |
| entangling-ZZ candidates, and the 30 separable-product candidates. The 60-candidate                                           | 1937 |
| pool is compared with 12 complete classical blocks and each 30-candidate stratum with                                         | 1938 |
| six complete classical blocks, always crossed with all five dimensions. Each classical                                        | 1939 |
| endpoint is the expectation over 5,000 <code>kernel_blocked</code> draws using SHA-256-derived                                | 1940 |
| seeds rooted at 20260729. This tests dependence on map stratum, not computational                                             | 1941 |
| advantage.                                                                                                                    | 1942 |
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| <b>4.18 Evaluation-choice factorial and shortcut sensitivity</b>                                                              | 1949 |
|                                                                                                                               | 1950 |
| The evaluation-choice SVC factorial uses all 360 q1000 runs and crosses: same-test                                            | 1951 |
| OOD versus disjoint ID-validation selection; fixed $C = 1$ versus train-CV $C$ ; cus-                                         | 1952 |
| tomary versus extended classical reference; and native versus equal-count budgets.                                            | 1953 |
| Fixed- $C$ predictions are recomputed with the deterministic controlled-protocol prepro-                                      | 1954 |
| cessing, kernel constructors, samples, and split roles; an integrity gate requires exact                                      | 1955 |
| candidate-split key identity with every archived controlled-protocol SVC summary                                              | 1956 |
| and the original ID-split hash salt. With the customary 30-candidate reference, equal                                         | 1957 |
| count samples six of twelve quantum blocks; with the extended reference, it samples                                           | 1958 |
| 12 of 23 classical blocks. The same 5,000 frozen block draws are paired across selection                                      | 1959 |
| and regularization cells. Sixteen fixed-case endpoints and their paired axis contrasts                                        | 1960 |
| are descriptive; they do not identify causal effects.                                                                         | 1961 |
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|                                                                                                                               | 1967 |
|                                                                                                                               | 1968 |
|                                                                                                                               | 1969 |
| The q1000 shortcut sensitivity uses the same design on all 270 network runs                                                   | 1970 |
| after removing features before the train-only embedding. UNSW-NB15 excludes                                                   | 1971 |
| <code>ct_src_dport_ltm</code> , <code>ct_dst_sport_ltm</code> , and <code>is_sm_ips_ports</code> ; ToN-IoT excludes           | 1972 |
| source/destination ports, all <code>proto_*</code> , <code>service</code> , and <code>service_*</code> fields. This removes 3 | 1973 |
| of 39 and 14 of 88 fields, respectively. The manuscript-facing contrast uses train-CV,                                        | 1974 |
|                                                                                                                               | 1975 |
|                                                                                                                               | 1976 |
|                                                                                                                               | 1977 |
|                                                                                                                               | 1978 |

P1', and 60-candidate block matching with paired draws. It is a port/protocol/service-field ablation, not a guarantee that every shortcut has been removed.

#### 4.19 Security fixed-case aggregation and conditional intervals

The effective replication axis within a scenario-group is q-split seed. Master seed is a design stratum rather than a replicate, some EMBER OOD pools recur across master seeds, and training sizes are nested. Audited OOD overlap is at most 1.2% for EMBER and 7.8% for network groups. Within each q-split cluster, model seeds are averaged within master seed, master seeds within size, and the three sizes with equal weight. The five resulting cluster means  $T_j$  give the group effect  $\bar{T}$  and 95% conditional cluster- $t$  interval

$$\bar{T} \pm t_{4,0.975} \frac{s_T}{\sqrt{5}}.$$

This few-cluster  $t$  construction follows the cluster-mean approach of Ibragimov and Müller [58]. The interval ( $n = 5$  q-split clusters) is conditional on the fixed benchmark pools and experimental design; inclusion of zero is not evidence of equivalence. A 9,999-draw BCa bootstrap over cluster means and leave-one-q-split calculations are secondary diagnostics, reported for all 16 primary cells in Supplementary Table 6. The five cluster values behind every interval are exposed in Supplementary Table 7.

Across groups, EMBER  $m1/m2$  are averaged within source dataset, the four UNSW-NB15 groups within UNSW-NB15, and the two ToN-IoT groups within ToN-IoT. These three source-dataset means receive equal weight. Leave-one-source-dataset-out ranges omit one of the three at a time. The eight scenario-groups and three datasets were not sampled from defined populations; therefore no population  $p$ -value or dataset-population confidence interval is reported.

## 4.20 Specification curve

The S1–S10 display is a prespecified cross-protocol sensitivity map over complete end-points from the earlier fixed- $C$  protocol and the controlled protocol. S1–S4 use the earlier fixed- $C$  protocol and S5–S10 the train-CV controlled protocol, so their boundary cannot isolate regularization. The rows remain in their prespecified order and are never sorted by effect. Individual-axis interpretation instead uses the evaluation-choice factorial above. All summaries average realizations within setting and group, then groups within the three source datasets, and finally source datasets with equal weight; no specification-level hypothesis test is performed.

## 4.21 Prospectively frozen external corroboration

Before acquiring TableShift data or inspecting any external result, we committed the tasks, sampling, preprocessing, candidate pools, estimands, uncertainty procedure, and outcome-category rule. The source implementation is pinned to commit `fca9429814703a07e3902d005d46563a207b7f0a` [51]. Three public tasks span distinct domains and published shift variables: College Scorecard (institution type), diabetes readmission (admission source), and ACS income (geographic region). TableShift’s `train`, `validation`, `id_test`, and `ood_test` roles are preserved; `ood_validation` is never loaded. College Scorecard is read from its public CC0 archive because the pinned downloader invokes authenticated Kaggle access; the pinned parser, schema, and domain split are unchanged. Source-file SHA-256 hashes and the container digest are recorded.

Each task has nested strata  $(n_{\text{train}}, n_{\text{val}}, n_{\text{ID}}, n_{\text{OOD}}) = (1000, 250, 250, 500)$  and  $(2000, 500, 500, 1000)$ . For seeds  $\{42, 123, 999, 7, 2024\}$ , source rows are ordered by SHA-256 of task, split, seed, and original position, giving deterministic label-blind samples. Gates verify source-role disjointness, size nesting, both classes, source-schema

2071 preservation, and at least 12 non-constant encoded features. All 30 task-size-seed  
2072 units passed.

2074 All transformations fit `train` only. Constant columns are removed; numeric vari-  
2075 ables receive median imputation; Boolean, categorical, and string variables receive  
2076 most-frequent imputation and one-hot encoding with unknown categories ignored.  
2077 MaxAbs scaling, TruncatedSVD at the five study dimensions, standardization, and  
2078 a training-fitted affine map that places training coordinates in  $[0, \pi]$  follow; held-out  
2079 coordinates may extrapolate beyond that range. The SVD random state is 20260729.  
2080 The security kernel and classifier pools are then reused.

2086 For each task, size, seed, and classifier, the validation-selected 60-candidate quan-  
2087 tum endpoint is compared with 5,000 `kernel_blocked` 60-candidate subsamples of the  
2088 115-candidate classical pool using seed 20260729. The primary effect is quantum minus  
2089 expected classical OOD balanced accuracy. The equal-budget OOD oracle selects on  
2090 `ood_test`; the weak-fixed oracle uses linear/RBF, fixes SVC at  $C = 1$ , and also selects  
2091 on the reported OOD labels. Protocol contraction is weak-fixed-oracle effect minus  
2092 controlled P1' effect.

2097 Seeds are averaged within task and size, nested sizes receive equal weight within  
2098 task, and the three fixed tasks receive equal weight. Conditional 95% intervals use 9,999  
2099 resamples of the five seed clusters within task while preserving nested sizes. Leave-  
2100 one-task-out ranges are descriptive. The intervals quantify deterministic-subsampling  
2101 variability conditional on three fixed tasks; they do not represent a population of tasks.

## 2107 4.22 Performance and uncertainty metrics

2109 Balanced accuracy on  $S_{\text{OOD}}$  is primary. Accuracy, macro and positive-class F1,  
2110 ROC AUC, and precision-recall AUC are recorded on every split. The GP analysis

additionally records log loss, Brier score, expected calibration error with 15 equal-width probability bins [59], and mean predictive entropy. No threshold or calibration parameter is fitted on OOD labels.

### 4.23 Geometry descriptors and association analyses

For positive-semidefinite  $K$  with eigenvalues  $\lambda_i$  and  $p_i = \lambda_i / \sum_j \lambda_j$ , effective rank is

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right)$$

[60]. For  $y \in \{-1, +1\}^n$  and doubly centered  $K_c$ , centered kernel-target alignment is

$$\text{KTA}(K, y) = \frac{\langle K_c, yy^\top \rangle_F}{n \|K_c\|_F}$$

[61, 62]. KTA is computed separately on training, ID, and OOD square blocks, alongside off-diagonal and cross-split similarities. OOD KTA reuses the same OOD labels as accuracy and is therefore a post hoc diagnostic, not a label-independent mechanism.

Within each run–dimension cell, Spearman correlations relate effective rank, OOD KTA, and KTA survival (OOD minus ID) to SVC OOD balanced accuracy for all, classical-only, and quantum-only candidates. Partial Spearman association residualizes ranked variables against family and log length/angle scale. For portability, a descriptive model bins log effective rank into ten training-dataset quantiles, learns mean OOD accuracy per bin from two source datasets, and ranks cells in the third; held-out Spearman correlation is reported once per source dataset.

Rank matching is observational. Within each run and dimension, every quantum candidate is first paired to the classical candidate with smallest absolute effective-rank difference; classical candidates may be reused. Rank ratio  $\max(r_Q, r_C) / \min(r_Q, r_C)$  is filtered at prespecified calipers 1.10, 1.25, 1.50, and 2.00, plus unfiltered reporting. A

sensitivity uses minimum-cost one-to-one assignment with absolute log-rank difference, leaving excess candidates unmatched. Counts, retained fractions, rank discrepancies, mean and median OOD differences, and the fraction favouring quantum are reported for every caliper. None of these analyses identifies a causal geometry or family effect.

#### 4.24 Repeated finite-shot perturbation

The finite-shot analysis and its out-of-sample PSD extension were each specified before their outcomes were computed. They fix one  $q1000$ , master-seed-42, q-split-42, model-seed-42 run per security group and, within each, the exact-statevector quantum SVC candidate selected by  $P1'$  on  $V_{ID}$ . Lexicographic candidate order resolves ties. Input `summary_v4.csv` SHA-256 hashes, chosen kernels, dimensions, and exact  $C$  values are stored in the frozen specifications.

For shots  $s \in \{128, 512, 2048, 8192\}$  and measurement replicates  $r = 0, \dots, 29$ , each exact fidelity  $p$  is sampled independently as  $\text{Binomial}(s, p)/s$ . A separate NumPy generator is used for train, ID-validation, ID-test, OOD-test, and OOD-square blocks. Its unsigned 64-bit seed is the first eight SHA-256 bytes of

```
ksf-v6-shots::::<kernel>::::<shots>::::<block>.
```

Python's process-randomized hash is not used.

Square train and OOD blocks are sampled on the upper triangle, mirrored, clipped to  $[0, 1]$ , and assigned unit diagonal; rectangular blocks receive binomial sampling and clipping. Three prespecified conditions use the same sampled blocks. The unprojected condition makes no PSD correction. The independent-square heuristic clips negative eigenvalues of the train and OOD square matrices separately while leaving every rectangular evaluation-train block unchanged; it is an algorithmic intervention for a precomputed SVC, not a coherent correction of one out-of-sample kernel.



The coherent Nyström condition eigendecomposes the sampled training matrix as  
 $K_T = U \text{diag}(\lambda) U^\top$  and retains  $\lambda_i > \tau$ , where  $\tau = \max\{10^{-12}, \epsilon n \max(1, \max_i |\lambda_i|)\}$ .  
For retained  $U_+$  and  $\lambda_+$ ,

$$K_T^+ = U_+ \text{diag}(\lambda_+) U_+^\top, \quad \Phi_E = K_{E,T} U_+ \text{diag}(\lambda_+)^{-1/2}.$$

The corrected evaluation–train and evaluation-square blocks are  $\Phi_E \Phi_T^\top$  and  $\Phi_E \Phi_E^\top$ ,  
respectively, with  $\Phi_T = U_+ \text{diag}(\lambda_+)^{1/2}$ . Thus the OOD square used for KTA is  
derived from the sampled OOD–train block rather than independently projected. In  
all three conditions,  $C$  is reselected by the same five-fold train-only procedure and  
the final SVC is refitted. Outputs include signed and absolute OOD change from the  
fixed exact endpoint, effective-rank ratio, OOD-KTA change, retained rank, eigen-  
value tolerance, negative-eigenvalue fraction, and Frobenius changes for every block.  
Within each fixed Gram matrix, medians and central 2.5–97.5% Monte Carlo inter-  
vals summarize 30 replicates. Across the eight fixed matrices, medians and ranges are  
descriptive.

For the v0.9 extension we export the hard target prediction of every perturbed  
SVC and compare it with the unchanged exact-statevector classical witnesses. For each  
replicate and classical tier we record self-disagreement from the exact quantum predic-  
tor, the zero-label interval, and realized accuracy and balanced-accuracy advantages  
after the integrity-gated label join. This deliberately asks whether quantum measure-  
ment perturbation creates behaviour absent from the existing classical reference; it  
does not give the classical family an additional noise or retuning budget. The model  
contains no device or correlated-noise process.

## 2255 4.25 Statistics and reproducibility

2256  
2257 No null-hypothesis test is used for the eight security groups, three source datasets,  
2258  
2259 or three external tasks. Every uncertainty object names its effective  $n$ , resampling  
2260  
2261 unit, and conditioning set: security intervals use five q-split clusters; external intervals  
2262  
2263 use five deterministic seed clusters per task; finite-shot intervals use 30 measurement  
2264  
2265 replicates conditional on one fixed Gram matrix. The fifteen security pipeline realiza-  
2266  
2267 tions are never treated as independent. Security intervals use the  $t_4$  reference and are  
2268  
2269 pointwise, not simultaneous; the five underlying cluster means are published. Inclusion  
2270  
2271 of zero is never interpreted as equivalence. No multiplicity adjustment was prespec-  
2272  
2273 ified. Because multiple intervals and diagnostics are displayed, they are interpreted  
2274  
2275 descriptively rather than as a familywise error-controlled decision procedure; the BCa  
2276  
2277 analysis with five clusters is retained only as a fragile sensitivity.

2275     Sample sizes were design- and compute-constrained and fixed before outcome anal-  
2276  
2277 ysis; no power calculation was used. No completed group, task, specification, budget  
2278  
2279 scheme, caliper, or finite-shot replicate was removed based on its result. Randomized  
2280  
2281 operations use recorded seeds; external sampling, ID division, and audit tie-breaking  
2282  
2283 use deterministic SHA-256 ordering. The test suite validates source-dataset mapping,  
2284  
2285 endpoint equality across analyses, candidate-budget coverage, matching uniqueness,  
2286  
2287 seed stability, exact-reference hashes, prediction/label separation, theorem endpoints  
2288  
2289 by exhaustive enumeration, MILP consistency, monotone partial-label contraction,  
2290  
2291 and complete Monte Carlo cells.

2290

2291

## 2292 4.26 Software environment

2293

2294 Experiments use Python 3.11/3.12, NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0,  
2295  
2296 Matplotlib 3.11.0, and Qiskit 2.3.1. The Windows-specific Conda snapshot and cross-  
2297  
2298 platform package specifications serve different archival purposes; neither is an exact  
2299  
2300 machine image of every full-scale run.

|                                                                                            |      |
|--------------------------------------------------------------------------------------------|------|
| <b>4.27 Use of generative AI</b>                                                           | 2301 |
|                                                                                            | 2302 |
| Large language models were used beyond copy-editing, and no model is an author.            | 2303 |
| OpenAI Codex was used during manuscript revision for language editing, code inspection,    | 2304 |
| analysis implementation, and consistency checking. Claude Code (Anthropic),                | 2305 |
| running the Claude Fable 5.1 model, was used in September 2026 for editorial restructuring | 2306 |
| of the Introduction, literature search and reference formatting, adaptation to             | 2307 |
| the journal format, and consistency checking. No model determined authorship, study        | 2308 |
| design, data handling, analysis decisions, or the acceptance of any scientific claim. The  | 2309 |
| authors inspected every generated change, verified all reported analyses and citations     | 2310 |
| against primary sources, and take full responsibility for the manuscript.                  | 2311 |
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| <b>Declarations</b>                                                                        | 2319 |
|                                                                                            | 2320 |
|                                                                                            | 2321 |
| <b>Ethics approval and consent to participate.</b> This work is a secondary computational  | 2322 |
| analysis of public benchmark releases. The authors recruited no participants,              | 2323 |
| administered no intervention, collected no new human data, and accessed no direct          | 2324 |
| identifiers. The cybersecurity benchmarks contain malware or network-flow records          | 2325 |
| rather than human-participant observations. For the TableShift tasks, ACS PUMS             | 2326 |
| records are public-use files with Census Bureau disclosure protection [63]; BRFSS survey   | 2327 |
| data and documentation are publicly downloadable from CDC [64]; the NHANES                 | 2328 |
| source protocols are documented by the NCHS Ethics Review Board [65]; and the              | 2329 |
| diabetes-readmission records are distributed through the public UCI repository [66].       | 2330 |
| College Scorecard supplies public institution-level records [67]. Raw source records       | 2331 |
| are not redistributed. This study falls outside the scope of institutional ethics review   | 2332 |
| because it consists exclusively of secondary analyses of publicly available, deidentified  | 2333 |
| datasets, without recruitment, interaction, intervention, access to direct identifiers, or | 2334 |
| any attempt at reidentification. Accordingly, ethics approval and additional informed      | 2335 |
| consent were not required. No ethics approval or exemption is claimed.                     | 2336 |
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2347 **Consent for publication.** Not applicable.  
2348  
2349 **Availability of data and materials.** The derived datasets supporting the con-  
2350  
2351 clusions of this article are available in the Zenodo repository, [https://doi.org/](https://doi.org/10.5281/zenodo.21776862)  
2352 10.5281/zenodo.21776862 (immutable v1.1.6 release). This study uses the public  
2353  
2354 EMBER [52], UNSW-NB15 [53], ToN-IoT [54], and TableShift [51] benchmarks,  
2355  
2356 which remain subject to their original access conditions; raw source records are  
2357  
2358 not redistributed. The archived release contains derived split definitions, deter-  
2359  
2360 ministic row-position hashes, shift constructions, experiment manifests, complete  
2361  
2362 aggregated results, target-prediction matrices, physically separated audit labels, evi-  
2363  
2364 dence frontiers, technical-failure records, and integrity manifests. The all-version  
2365  
2366 concept DOI is <https://doi.org/10.5281/zenodo.19147649>; the exact v1.1.5 prede-  
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2368 cessor is <https://doi.org/10.5281/zenodo.21764577>, the exact v1.1.4 predecessor is  
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2370 <https://doi.org/10.5281/zenodo.21759058>, and the exact v0.8.0 predecessor is <https://doi.org/10.5281/zenodo.21717074> [68].  
2371  
2372 Code for constructing the splits, running the kernel experiments, reproduc-  
2373  
2374 ing every analysis, and regenerating every table and figure is available as fol-  
2375  
2376 lows. Project name: kernel\_shift\_framework. Project home page: [https://github.com/](https://github.com/roberto-fernandez-barrios/kernel_shift_framework)  
2377  
2378 roberto-fernandez-barrios/kernel\_shift\_framework. Archived version: [https://doi.org/](https://doi.org/10.5281/zenodo.21776862)  
2379  
2380 10.5281/zenodo.21776862. Operating system(s): platform independent; a Windows-  
2381  
2382 specific Conda snapshot and platform-neutral package specifications document the  
2383  
2384 dependencies, and neither is claimed as an exact machine image of every full-scale  
2385  
2386 run. Programming language: Python. Other requirements: Python 3.11 or later with  
2387  
2388 NumPy, pandas, scikit-learn, Matplotlib, and Qiskit at the versions listed in the  
2389  
2390 archived lock files. License: BSD-3-Clause. Any restrictions to use by non-academics:  
2391  
2392 none.

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2390  
2391 interests.  
2392

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|                                   | curation, formal analysis, visualization, investigation, writing—original draft, and                                                               | 2396 |
|                                   | writing—review and editing. I.P.-L.: supervision, validation, and writing—review and                                                               | 2397 |
|                                   | editing. A.G.-S.: validation, project administration, and writing—review and editing.                                                              | 2398 |
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|                                   | approved the final manuscript.                                                                                                                     | 2400 |
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| <b>Supplementary information.</b> | Additional file 1: Supplementary Information                                                                                                       | 2408 |
|                                   | (PDF). Extended results, complete sensitivity tables, the prospective Gate-2 repli-                                                                | 2409 |
|                                   | cation tables against the prespecified classical-kernel reference family, geometry and                                                             | 2410 |
|                                   | finite-shot diagnostics, and the machine-readable result map supporting the main                                                                   | 2411 |
|                                   | article.                                                                                                                                           | 2412 |
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| <b>References</b>                 |                                                                                                                                                    | 2417 |
|                                   |                                                                                                                                                    | 2418 |
|                                   |                                                                                                                                                    | 2419 |
| [1]                               | Havlíček V, Córcoles AD, Temme K, Harrow AW, Kandala A, Chow JM,                                                                                   | 2420 |
|                                   | et al. Supervised Learning with Quantum-Enhanced Feature Spaces. <i>Nature</i> .                                                                   | 2421 |
|                                   | 2019;567:209–212. <a href="https://doi.org/10.1038/s41586-019-0980-2">https://doi.org/10.1038/s41586-019-0980-2</a> .                              | 2422 |
|                                   |                                                                                                                                                    | 2423 |
|                                   |                                                                                                                                                    | 2424 |
|                                   |                                                                                                                                                    | 2425 |
| [2]                               | Schuld M, Killoran N. Quantum Machine Learning in Feature Hilbert                                                                                  | 2426 |
|                                   | Spaces. <i>Physical Review Letters</i> . 2019;122(4):040504. <a href="https://doi.org/10.1103/PhysRevLett.122.040504">https://doi.org/10.1103/</a> | 2427 |
|                                   | <i>PhysRevLett</i> .122.040504.                                                                                                                    | 2428 |
|                                   |                                                                                                                                                    | 2429 |
|                                   |                                                                                                                                                    | 2430 |
|                                   |                                                                                                                                                    | 2431 |
| [3]                               | Schuld M. Supervised Quantum Machine Learning Models Are Kernel Methods.                                                                           | 2432 |
|                                   | arXiv. 2021;abs/2101.11020. <a href="https://doi.org/10.48550/arXiv.2101.11020">https://doi.org/10.48550/arXiv.2101.11020</a> .                    | 2433 |
|                                   |                                                                                                                                                    | 2434 |
|                                   |                                                                                                                                                    | 2435 |
|                                   |                                                                                                                                                    | 2436 |
|                                   |                                                                                                                                                    | 2437 |
|                                   |                                                                                                                                                    | 2438 |

- 2439 [4] Bowles J, Ahmed S, Schuld M. Better than Classical? The Subtle Art of  
2440 Benchmarking Quantum Machine Learning Models. arXiv. 2024;abs/2403.07059.  
2441 <https://doi.org/10.48550/arXiv.2403.07059>.  
2442  
2443  
2444
- 2445 [5] Slattery L, Shaydulin R, Chakrabarti S, Pistoia M, Khairy S, Wild SM. Numerical  
2446 Evidence against Advantage with Quantum Fidelity Kernels on Classical Data.  
2447 Physical Review A. 2023;107(6):062417. [https://doi.org/10.1103/PhysRevA.107.](https://doi.org/10.1103/PhysRevA.107.062417)  
2448 062417.  
2449  
2450  
2451
- 2452 [6] Schnabel J, Roth M. Quantum Kernel Methods under Scrutiny: A Benchmark-  
2453 ing Study. Quantum Machine Intelligence. 2025;7:58. [https://doi.org/10.1007/](https://doi.org/10.1007/s42484-025-00273-5)  
2454 s42484-025-00273-5.  
2455  
2456  
2457
- 2458 [7] Gillani SA, Baig MSA, Shah SM, Ali A, Siddiqui H. How Quantum Is the  
2459 Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution  
2460 Audit of Quantum Machine Learning for Network Intrusion Detection. arXiv.  
2461 2026;abs/2608.18155. <https://doi.org/10.48550/arXiv.2608.18155>.  
2462  
2463  
2464
- 2465 [8] Shen J. Quantum Kernels and the Cross-Section of Stock Returns: Anatomy of  
2466 a Vanishing Advantage. arXiv. 2026;abs/2607.20168. [https://doi.org/10.48550/](https://doi.org/10.48550/arXiv.2607.20168)  
2467 arXiv.2607.20168.  
2468  
2469  
2470
- 2471 [9] Gulrajani I, Lopez-Paz D. In Search of Lost Domain Generalization. In: Inter-  
2472 national Conference on Learning Representations. International Conference on  
2473 Learning Representations; 2021. Available from: [https://openreview.net/forum?](https://openreview.net/forum?id=lQdXeXDoWtI)  
2474 id=lQdXeXDoWtI.  
2475  
2476  
2477
- 2478 [10] Koh PW, Sagawa S, Marklund H, Xie SM, Zhang M, Balsubramani A, et al.  
2479 WILDS: A Benchmark of In-the-Wild Distribution Shifts. In: Proceedings of  
2480 the 38th International Conference on Machine Learning. vol. 139 of Proceedings  
2481  
2482  
2483  
2484

- of Machine Learning Research. International Conference on Machine Learning. 2485  
Cambridge, MA, USA: PMLR; 2021. p. 5637–5664. Available from: [https://](https://proceedings.mlr.press/v139/koh21a.html) 2486  
[proceedings.mlr.press/v139/koh21a.html](https://proceedings.mlr.press/v139/koh21a.html). 2487  
2488  
2489  
2490
- [11] Yao H, Choi C, Cao B, Lee Y, Koh PW, Finn C. Wild-Time: A Benchmark of 2491  
In-the-Wild Distribution Shift over Time. In: Advances in Neural Information 2492  
Processing Systems. Neural Information Processing Systems Foundation; 2022. 2493  
2494  
2495  
2496  
2497  
2498
- [12] Zhou K, Liu Z, Qiao Y, Xiang T, Loy CC. Domain Generalization: A 2499  
Survey. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2500  
2023;45(4):4396–4415. <https://doi.org/10.1109/TPAMI.2022.3195549>. 2501  
2502  
2503
- [13] Huang HY, Broughton M, Mohseni M, Babbush R, Boixo S, Neven H, et al. Power 2504  
of Data in Quantum Machine Learning. Nature Communications. 2021;12:2631. 2505  
<https://doi.org/10.1038/s41467-021-22539-9>. 2506  
2507  
2508  
2509
- [14] Schreiber FJ, Eisert J, Meyer JJ. Classical Surrogates for Quantum Learning 2510  
Models. Physical Review Letters. 2023;131:100803. [https://doi.org/10.1103/](https://doi.org/10.1103/PhysRevLett.131.100803) 2511  
[PhysRevLett.131.100803](https://doi.org/10.1103/PhysRevLett.131.100803). 2512  
2513  
2514  
2515
- [15] Tekeli E. The ZZ feature map induces a signless Laplacian metric: a closed-form 2516  
classical surrogate for quantum kernel regression. arXiv. 2026;abs/2608.29422. 2517  
<https://doi.org/10.48550/arXiv.2608.29422>. 2518  
2519  
2520  
2521
- [16] Thabet S, Monbroussou L, Mamon EZ, Landman J. When Quantum and Classical 2522  
Models Disagree: Learning Beyond Minimum Norm Least Square. npj Quantum 2523  
Information. 2026;12:81. <https://doi.org/10.1038/s41534-026-01217-y>. 2524  
2525  
2526  
2527
- [17] Manski CF. Partial Identification of Probability Distributions. Springer Series in 2528  
Statistics. New York, NY: Springer; 2003. 2529  
2530

- 2531 [18] Molinari F. Microeconometrics with Partial Identification. In: Durlauf SN,  
2532 Hansen LP, Heckman JJ, Matzkin RL, editors. Handbook of Econometrics,  
2533 Volume 7A. Elsevier; 2020. p. 355–486.  
2534  
2535
- 2536 [19] Polo FM, Maity S, Yurochkin M, Banerjee M, Sun Y. Weak Supervision Perfor-  
2537 mance Evaluation via Partial Identification. In: Advances in Neural Information  
2538 Processing Systems. vol. 37; 2024. .  
2539  
2540  
2541
- 2542 [20] Katariya N, Iyer A, Sarawagi S. Active Evaluation of Classifiers on Large  
2543 Datasets. In: 2012 IEEE 12th International Conference on Data Mining. IEEE;  
2544 2012. p. 329–338.  
2545  
2546  
2547
- 2548 [21] Kossen J, Farquhar S, Gal Y, Rainforth T. Active Testing: Sample-Efficient Model  
2549 Evaluation. In: Proceedings of the 38th International Conference on Machine  
2550 Learning. vol. 139 of Proceedings of Machine Learning Research. PMLR; 2021. p.  
2551 5753–5763. Available from: <https://proceedings.mlr.press/v139/kossen21a.html>.  
2552  
2553  
2554  
2555
- 2556 [22] Kossen J, Farquhar S, Gal Y, Rainforth T. Active Surrogate Estima-  
2557 tors: An Active Learning Approach to Label-Efficient Model Evaluation. In:  
2558 Advances in Neural Information Processing Systems. vol. 35; 2022. p. 24557–  
2559 24570. Available from: [https://proceedings.neurips.cc/paper\\_files/paper/2022/  
2560 hash/9b9cfd5428153ccfbd4ba34b7e007305-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/9b9cfd5428153ccfbd4ba34b7e007305-Abstract-Conference.html).  
2561  
2562  
2563  
2564
- 2565 [23] Ashouritaklimi K, Kilian V, Huang D, Rainforth T, Caron F. Prediction-Powered  
2566 Active Testing. arXiv. 2026;abs/2607.08347. [https://doi.org/10.48550/arXiv.  
2567 2607.08347](https://doi.org/10.48550/arXiv.2607.08347).  
2568  
2569  
2570
- 2571 [24] Okanovic P, Kirsch A, Kasper J, Hoeffler T, Krause A, Gürel NM. All Models Are  
2572 Wrong, Some Are Useful: Model Selection with Limited Labels. In: Proceedings  
2573 of The 28th International Conference on Artificial Intelligence and Statistics. vol.  
2574  
2575  
2576



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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 258 of Proceedings of Machine Learning Research. PMLR; 2025. p. 2035–2043.                                                                                                                                       | 2577 |
| Available from: <a href="https://proceedings.mlr.press/v258/okanovic25a.html">https://proceedings.mlr.press/v258/okanovic25a.html</a> .                                                                          | 2578 |
|                                                                                                                                                                                                                  | 2579 |
|                                                                                                                                                                                                                  | 2580 |
| [25] Shanmugam D, Sadhuka S, Raghavan M, Gutttag J, Berger B, Pierson                                                                                                                                            | 2581 |
| E. Evaluating Multiple Models Using Labeled and Unlabeled Data. In:                                                                                                                                              | 2582 |
| Advances in Neural Information Processing Systems. vol. 38; 2025. p. 30612–                                                                                                                                      | 2583 |
| 30648. Available from: <a href="https://proceedings.neurips.cc/paper_files/paper/2025/hash/2c0e78cc177dfeca260ef990a8c99209-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2025/</a> | 2584 |
| <a href="https://proceedings.neurips.cc/paper_files/paper/2025/hash/2c0e78cc177dfeca260ef990a8c99209-Abstract-Conference.html">hash/2c0e78cc177dfeca260ef990a8c99209-Abstract-Conference.html</a> .              | 2585 |
|                                                                                                                                                                                                                  | 2586 |
|                                                                                                                                                                                                                  | 2587 |
|                                                                                                                                                                                                                  | 2588 |
|                                                                                                                                                                                                                  | 2589 |
| [26] Garg S, Balakrishnan S, Lipton ZC, Neyshabur B, Sedghi H. Leveraging Unla-                                                                                                                                  | 2590 |
| beled Data to Predict Out-of-Distribution Performance. In: International Con-                                                                                                                                    | 2591 |
| ference on Learning Representations; 2022. Available from: <a href="https://openreview.net/forum?id=o_HsiMPYh_x">https://openreview.</a>                                                                         | 2592 |
| <a href="https://openreview.net/forum?id=o_HsiMPYh_x">net/forum?id=o_HsiMPYh_x</a> .                                                                                                                             | 2593 |
|                                                                                                                                                                                                                  | 2594 |
|                                                                                                                                                                                                                  | 2595 |
|                                                                                                                                                                                                                  | 2596 |
|                                                                                                                                                                                                                  | 2597 |
| [27] Rosenfeld E, Garg S. (Almost) Provable Error Bounds Under Distribution Shift                                                                                                                                | 2598 |
| via Disagreement Discrepancy. In: Advances in Neural Information Processing                                                                                                                                      | 2599 |
| Systems. vol. 36; 2023. p. 28761–28784.                                                                                                                                                                          | 2600 |
|                                                                                                                                                                                                                  | 2601 |
|                                                                                                                                                                                                                  | 2602 |
|                                                                                                                                                                                                                  | 2603 |
| [28] Mishra A, Liu A. ODD: Overlap-Aware Estimation of Model Performance under                                                                                                                                   | 2604 |
| Distribution Shift. In: Proceedings of the 41st Conference on Uncertainty in                                                                                                                                     | 2605 |
| Artificial Intelligence. vol. 286 of Proceedings of Machine Learning Research.                                                                                                                                   | 2606 |
| PMLR; 2025. p. 3031–3047. Available from: <a href="https://proceedings.mlr.press/v286/mishra25a.html">https://proceedings.mlr.press/v286/</a>                                                                    | 2607 |
| <a href="https://proceedings.mlr.press/v286/mishra25a.html">mishra25a.html</a> .                                                                                                                                 | 2608 |
|                                                                                                                                                                                                                  | 2609 |
|                                                                                                                                                                                                                  | 2610 |
|                                                                                                                                                                                                                  | 2611 |
|                                                                                                                                                                                                                  | 2612 |
| [29] Han W, Qu L. The Label Complexity of Class-Conditional Coverage under Distri-                                                                                                                               | 2613 |
| bution Shift. arXiv. 2026;abs/2607.18088. <a href="https://doi.org/10.48550/arXiv.2607.18088">https://doi.org/10.48550/arXiv.2607.</a>                                                                           | 2614 |
| 18088.                                                                                                                                                                                                           | 2615 |
|                                                                                                                                                                                                                  | 2616 |
|                                                                                                                                                                                                                  | 2617 |
|                                                                                                                                                                                                                  | 2618 |
| [30] Madani O, Pennock DM, Flake GW. Co-Validation: Using Model Dis-                                                                                                                                             | 2619 |
| agreement on Unlabeled Data to Validate Classification Algorithms. In:                                                                                                                                           | 2620 |
|                                                                                                                                                                                                                  | 2621 |
|                                                                                                                                                                                                                  | 2622 |

Advances in Neural Information Processing Systems. vol. 17; 2004. p. 873–880. Available from: [https://proceedings.neurips.cc/paper\\_files/paper/2004/hash/92f54963fc39a9d87c2253186808ea61-Abstract.html](https://proceedings.neurips.cc/paper_files/paper/2004/hash/92f54963fc39a9d87c2253186808ea61-Abstract.html).

[31] Musgrove D, Radtke A, Haddad T. Discordant Pair Analysis for Sample Efficient Model Evaluation. *Scientific Reports*. 2023;13:20987. <https://doi.org/10.1038/s41598-023-48017-4>.

[32] Balsubramani A, Freund Y. Optimally Combining Classifiers Using Unlabeled Data. In: *Proceedings of the 28th Conference on Learning Theory*. vol. 40 of *Proceedings of Machine Learning Research*. PMLR; 2015. p. 211–225. Available from: <https://proceedings.mlr.press/v40/Balsubramani15.html>.

[33] Fernández-Barrios R, Pastor-López I, González-Santocildes A, García Bringas P. Conditional validity of quantum event classifiers under collider systematics and quantum estimation uncertainty. *arXiv*. 2026;abs/2609.02781. <https://doi.org/10.48550/arXiv.2609.02781>.

[34] Abreu D, Esteve Rothenberg C, Abelém A. QML-IDS: Quantum Machine Learning Intrusion Detection System. In: *2024 IEEE Symposium on Computers and Communications (ISCC)*. IEEE Computer Society. Paris, France: IEEE; 2024. p. 1–6.

[35] Abreu D, Moura D, Esteve Rothenberg C, Abelém A. QuantumNetSec: Quantum Machine Learning for Network Security. *International Journal of Network Management*. 2025;35(4):e70018. <https://doi.org/10.1002/nem.70018>.

[36] Bellante A, Fioravanti T, Carminati M, Zanero S, Luongo A. Evaluating the Potential of Quantum Machine Learning in Cybersecurity: A Case-Study on PCA-Based Intrusion Detection Systems. *Computers & Security*. 2025;154:104341.

|                                                                                                                                         |      |
|-----------------------------------------------------------------------------------------------------------------------------------------|------|
| <a href="https://doi.org/10.1016/j.cose.2025.104341">https://doi.org/10.1016/j.cose.2025.104341</a> .                                   | 2669 |
|                                                                                                                                         | 2670 |
| [37] Bazinet M, Zantedeschi V, Germain P. Bound to Disagree: Generalization                                                             | 2671 |
| Bounds via Certifiable Surrogates. arXiv. 2026;abs/2602.23128. <a href="https://doi.org/10.48550/arXiv.2602.23128">https://doi.org/</a> | 2672 |
| 10.48550/arXiv.2602.23128.                                                                                                              | 2673 |
|                                                                                                                                         | 2674 |
|                                                                                                                                         | 2675 |
|                                                                                                                                         | 2676 |
| [38] Quiñonero-Candela J, Sugiyama M, Schwaighofer A, Lawrence ND, editors.                                                             | 2677 |
| Dataset Shift in Machine Learning. Cambridge, MA: MIT Press; 2008.                                                                      | 2678 |
|                                                                                                                                         | 2679 |
|                                                                                                                                         | 2680 |
| [39] Shaydulin R, Wild SM. Importance of Kernel Bandwidth in Quantum Machine                                                            | 2681 |
| Learning. Physical Review A. 2022;106(4):042407. <a href="https://doi.org/10.1103/PhysRevA.106.042407">https://doi.org/10.1103/</a>     | 2682 |
| PhysRevA.106.042407.                                                                                                                    | 2683 |
|                                                                                                                                         | 2684 |
|                                                                                                                                         | 2685 |
|                                                                                                                                         | 2686 |
| [40] Canatar A, Peters E, Pehlevan C, Wild SM, Shaydulin R. Bandwidth Enables                                                           | 2687 |
| Generalization in Quantum Kernel Models. Transactions on Machine Learning                                                               | 2688 |
| Research. 2023;.                                                                                                                        | 2689 |
|                                                                                                                                         | 2690 |
|                                                                                                                                         | 2691 |
|                                                                                                                                         | 2692 |
| [41] Kübler JM, Buchholz S, Schölkopf B. The Inductive Bias of Quantum Kernels. In:                                                     | 2693 |
| Advances in Neural Information Processing Systems (NeurIPS). vol. 34. Neural                                                            | 2694 |
| Information Processing Systems Foundation; 2021. p. 12661–12673.                                                                        | 2695 |
|                                                                                                                                         | 2696 |
|                                                                                                                                         | 2697 |
|                                                                                                                                         | 2698 |
| [42] Thanasilp S, Wang S, Cerezo M, Holmes Z. Exponential Concentration in                                                              | 2699 |
| Quantum Kernel Methods. Nature Communications. 2024;15:5200.                                                                            | 2700 |
|                                                                                                                                         | 2701 |
|                                                                                                                                         | 2702 |
| [43] Wang X, Du Y, Luo Y, Tao D. Towards Understanding the Power of Quan-                                                               | 2703 |
| tum Kernels in the NISQ Era. Quantum. 2021;5:531. <a href="https://doi.org/10.22331/q-2021-08-30-531">https://doi.org/10.22331/</a>     | 2704 |
| q-2021-08-30-531.                                                                                                                       | 2705 |
|                                                                                                                                         | 2706 |
|                                                                                                                                         | 2707 |
|                                                                                                                                         | 2708 |
| [44] Gentinetta G, Thomsen A, Sutter D, Woerner S. The Complexity of Quantum                                                            | 2709 |
| Support Vector Machines. Quantum. 2024;8:1225. <a href="https://doi.org/10.22331/q-2024-01-11-1225">https://doi.org/10.22331/</a>       | 2710 |
| q-2024-01-11-1225.                                                                                                                      | 2711 |
|                                                                                                                                         | 2712 |
|                                                                                                                                         | 2713 |
|                                                                                                                                         | 2714 |

- 2715 [45] Miroszewski A. Adaptive Measurement Allocation for Learning Kernelized SVMs  
2716 Under Noisy Observations. arXiv. 2026;abs/2605.22275. [https://doi.org/10.](https://doi.org/10.48550/arXiv.2605.22275)  
2717 [48550/arXiv.2605.22275](https://doi.org/10.48550/arXiv.2605.22275).  
2718  
2719  
2720  
2721 [46] Agliardi G, Cortiana G, Dekusar A, Ghosh K, Mohseni N, O'Meara C, et al.  
2722 Mitigating Exponential Concentration in Covariant Quantum Kernels for Sub-  
2723 space and Real-World Data. npj Quantum Information. 2026;12:12. [https:](https://doi.org/10.1038/s41534-025-01154-2)  
2724 [//doi.org/10.1038/s41534-025-01154-2](https://doi.org/10.1038/s41534-025-01154-2).  
2725  
2726  
2727  
2728 [47] Liu Y, Arunachalam S, Temme K. A Rigorous and Robust Quantum Speed-  
2729 Up in Supervised Machine Learning. Nature Physics. 2021;17:1013–1017. [https:](https://doi.org/10.1038/s41567-021-01287-z)  
2730 [//doi.org/10.1038/s41567-021-01287-z](https://doi.org/10.1038/s41567-021-01287-z).  
2731  
2732  
2733  
2734 [48] Jerbi S, Gyurik C, Marshall SC, Molteni R, Dunjko V. Shadows of Quantum  
2735 Machine Learning. Nature Communications. 2024;15:5676. [https://doi.org/10.](https://doi.org/10.1038/s41467-024-49877-8)  
2736 [1038/s41467-024-49877-8](https://doi.org/10.1038/s41467-024-49877-8).  
2737  
2738  
2739  
2740 [49] Engelen G, Rimmer V, Joosen W. Troubleshooting an Intrusion Detection  
2741 Dataset: The CICIDS2017 Case Study. In: 2021 IEEE Security and Privacy Work-  
2742 shops (SPW). IEEE Computer Society. San Francisco, CA, USA: IEEE; 2021. p.  
2743 7–12.  
2744  
2745  
2746  
2747 [50] Kakavand S, Strohmeyer C, Schlotter M. Benchmarking Quantum Kernel Sup-  
2748 port Vector Machines Against Classical Baselines on Tabular Data: A Rigorous  
2749 Empirical Study with Hardware Validation. arXiv. 2026;abs/2604.18837. [https:](https://doi.org/10.48550/arXiv.2604.18837)  
2750 [//doi.org/10.48550/arXiv.2604.18837](https://doi.org/10.48550/arXiv.2604.18837).  
2751  
2752  
2753  
2754  
2755 [51] Gardner J, Popovic Z, Schmidt L. Benchmarking Distribution Shift in Tabular  
2756 Data with TableShift. In: Advances in Neural Information Processing Systems.  
2757 vol. 36. Neural Information Processing Systems Foundation; 2023. .  
2758  
2759  
2760

- [52] Anderson HS, Roth P. EMBER: An Open Dataset for Training Static PE Malware Machine Learning Models. arXiv. 2018;abs/1804.04637. 2761  
2762  
2763  
2764
- [53] Moustafa N, Slay J. UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems (UNSW-NB15 Network Data Set),. In: 2015 Military Communications and Information Systems Conference (MilCIS). IEEE; 2015. p. 1–6. 2765  
2766  
2767  
2768  
2769  
2770  
2771  
2772
- [54] Alsaedi A, Moustafa N, Tari Z, Mahmood A, Anwar A. TON.IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems. IEEE Access. 2020;8:165130–165150. <https://doi.org/10.1109/ACCESS.2020.3022862>. 2773  
2774  
2775  
2776  
2777  
2778  
2779
- [55] Cortes C, Vapnik V. Support-Vector Networks. Machine Learning. 1995;20(3):273–297. <https://doi.org/10.1007/BF00994018>. 2780  
2781  
2782  
2783
- [56] Rasmussen CE, Williams CKI. Gaussian Processes for Machine Learning. Cambridge, MA: MIT Press; 2006. 2784  
2785  
2786  
2787
- [57] Javadi-Abhari A, Treinish M, Krsulich K, Wood CJ, Lishman J, Gacon J, et al. Quantum Computing with Qiskit. arXiv. 2024;abs/2405.08810. <https://doi.org/10.48550/arXiv.2405.08810>. 2788  
2789  
2790  
2791  
2792  
2793
- [58] Ibragimov R, Müller UK. Inference with Few Heterogeneous Clusters. The Review of Economics and Statistics. 2016;98(1):83–96. [https://doi.org/10.1162/REST\\_a\\_00545](https://doi.org/10.1162/REST_a_00545). 2794  
2795  
2796  
2797  
2798  
2799
- [59] Guo C, Pleiss G, Sun Y, Weinberger KQ. On Calibration of Modern Neural Networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML). vol. 70 of PMLR. International Conference on Machine Learning; 2017. p. 1321–1330. 2800  
2801  
2802  
2803  
2804  
2805  
2806

2807 [60] Roy O, Vetterli M. The Effective Rank: A Measure of Effective Dimensionality.  
2808  
2809 In: 2007 15th European Signal Processing Conference (EUSIPCO). EURASIP;  
2810 2007. p. 606–610.  
2811

2812 [61] Cristianini N, Shawe-Taylor J, Elisseeff A, Kandola J. On Kernel-Target Align-  
2813 ment. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 14.  
2814  
2815 Neural Information Processing Systems Foundation; 2001. .  
2816  
2817

2818 [62] Cortes C, Mohri M, Rostamizadeh A. Algorithms for Learning Kernels Based on  
2819 Centered Alignment. Journal of Machine Learning Research. 2012;13(28):795–  
2820  
2821 828.  
2822  
2823

2824 [63] United States Census Bureau. Public Use Microdata Sample (PUMS). Accessed  
2825 2 August 2026. <https://www.census.gov/programs-surveys/acs/microdata.html>.  
2826  
2827

2828 [64] Centers for Disease Control and Prevention. BRFSS Frequently Asked Questions.  
2829 Accessed 2 August 2026. [https://www.cdc.gov/brfss/about/brfss\\_faq.htm](https://www.cdc.gov/brfss/about/brfss_faq.htm).  
2830  
2831

2832 [65] National Center for Health Statistics. Ethics Review Board Approval: National  
2833 Health and Nutrition Examination Survey. Accessed 2 August 2026. [https://](https://www.cdc.gov/nchs/nhanes/about/erb.html)  
2834  
2835 [www.cdc.gov/nchs/nhanes/about/erb.html](https://www.cdc.gov/nchs/nhanes/about/erb.html).  
2836  
2837

2838 [66] UCI Machine Learning Repository. Diabetes 130-US Hospitals for Years  
2839 1999–2008. Available from: [https://archive.ics.uci.edu/dataset/296/](https://archive.ics.uci.edu/dataset/296/diabetes-130-us-hospitals-for-years-1999-2008)  
2840  
2841 [diabetes-130-us-hospitals-for-years-1999-2008](https://archive.ics.uci.edu/dataset/296/diabetes-130-us-hospitals-for-years-1999-2008).  
2842  
2843

2844 [67] United States Department of Education. College Scorecard Data. Accessed 2  
2845 August 2026. <https://collegescorecard.ed.gov/data/>.  
2846  
2847

2848 [68] Fernández-Barrios R, Pastor-López I, González-Santocildes A, García Bringas  
2849 P. Evaluation Choices Shape Apparent Quantum Kernel Robustness under  
2850  
2851  
2852

Distribution Shift. Zenodo. Available from: <https://doi.org/10.5281/zenodo.21717074>. 2853  
2854  
2855  
2856  
2857  
2858  
2859  
2860  
2861  
2862  
2863  
2864  
2865  
2866  
2867  
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